

ALLPLAN 2021

Views and Sections

Steps to Success

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Welcome!

This step-by-step guide shows you how to create general arrangement drawings and reinforcement drawings based on views and sections that automatically update themselves so that you can tap the full potential of integrated design – from building model to reinforcement.

Five easy-to-follow steps guide you from building model generation to creation of general arrangement drawings and reinforcement drawings to layout output and data exchange.

Based on a simple and clear training project, this guide describes each step in detail so that you can follow quickly. As the training project contains the finished data for each step, you can get started wherever you want.

The appendix shows you how to work without links while creating general arrangement drawings and reinforcement drawings based on views and sections that automatically update themselves.

Have fun with this guide! We wish you every success!

Basics

In Allplan 2021, you can create general arrangement drawings and reinforcement drawings in two different ways.

You can work with or without a model. When working with a model, you use associative views and sections.

Note: You can use both the tools in the **Reinforcement Views** task area and the tools in the **Sections** task area to create model-based general arrangement drawings and reinforcement drawings. However, you cannot mix the tools from these two task areas. This guide describes how to use the tools in the **Sections** task area.

For whom is this guide intended?

This guide helps you getting started if you have created the general arrangement and reinforcement in the  **Reinforcement Views** task area until now and it is your first time working with the tools of the **Sections** task area in Allplan 2021. In addition, you will learn how to use existing reinforcement again (see "Using reinforcement again" on page 50).

You cannot create reinforcement with a 3D model based on a 2D general arrangement drawing in Allplan 2021. If you have a complex component that you do not want to model in detail, you can still create a three-dimensional model of the reinforcement by creating an auxiliary 3D solid. For example, use the  **Box** tool to create a box that is parallel to the coordinate planes and that has the dimensions of the component that you want to reinforce. To find out how to do this, see "Reinforcement based on an auxiliary 3D solid (see "Reinforcement based on an auxiliary 3D object" on page 58)".

If you work in 2D all the time, this guide gives you a quick and practical introduction to working in 3D, which has numerous advantages.

Overview of steps

Step 1 – building model

- Organization in the building structure
- Layers and print sets
- Area styles and drawing types
- Textures for surfaces

Step 2 – general arrangement views

- Structuring and selecting drawing files
- References and view options
- Creating associative sections

Step 3 – reinforcement

- Creating reinforcement with a 3D model
- Selecting drawing files for modifying data
- Using reinforcement again
- Managing reference drawing files
- 2D general arrangement and 3D reinforcement

Step 4 – layout output

- Assembling layouts
- Layout without model data

Step 5 – data exchange

- Exporting drawing files
- Exporting layouts

Appendix – working without links

- Organization in the building structure
- View options
- Copying and converting a part of the floor plan
- Using changed data of the floor plan
- Using reinforcement again
- Exchanging data

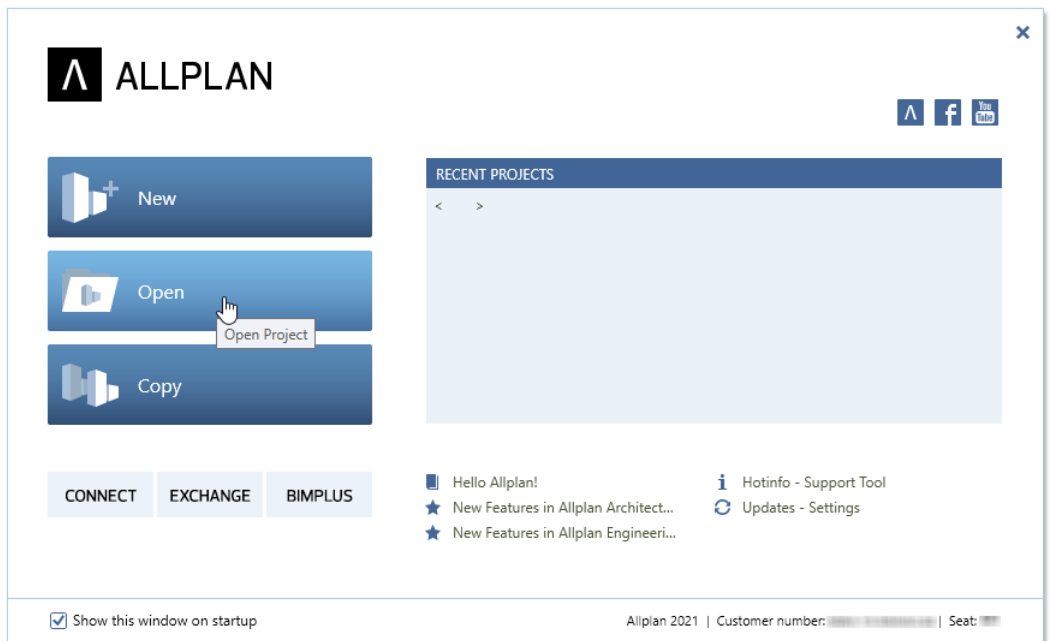
Importing the compressed project with the training data

We prepared a training project for you so that you can get started immediately and create general arrangement drawings and reinforcement drawings. Start by importing this training project into Allplan 2021.

You can download the compressed project from Allplan Connect (<https://connect.allplan.com>) on the internet. You can find the data in the **Training** area. Go to **Documentation – Step by Step**.

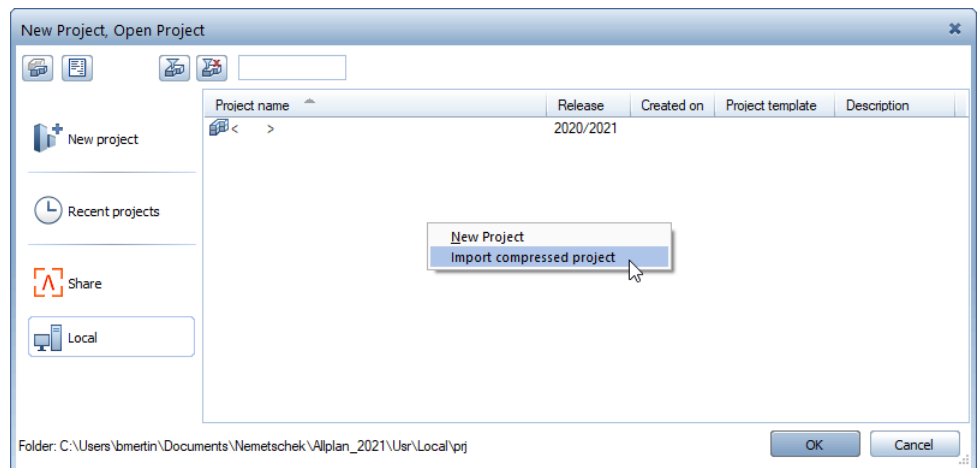
To start Allplan and to import the training project

- Allplan 2021 must be installed, registered, and configured. After having installed Allplan, start Allplan at least once to check whether it works properly.
 - You downloaded the compressed project from Allplan Connect (<https://connect.allplan.com>) and saved it into a folder of your choice.
- 1 Start  **Allplan 2021**. Click **Open** on the **welcome screen**.



If you have turned off the welcome screen, click  **New Project, Open Project...** on the Quick Access Toolbar.

- 2 Right-click in the empty area of the project list and select **Import compressed project** on the shortcut menu.



- 3 Select the downloaded ZIP file in the **Open** dialog box.

Note: You can also drag the downloaded ZIP file from File Explorer into the area where the projects are listed in the **New Project, Open Project** dialog box.

A project that you backed up to a ZIP file by using the **Services** application can also be imported directly to Allplan by drag-and-drop editing.

- 4 You can see the **New Project** dialog box, where you can adjust the **Location** and **Project name**. Click **OK** to confirm the dialog box.

The imported project is selected.

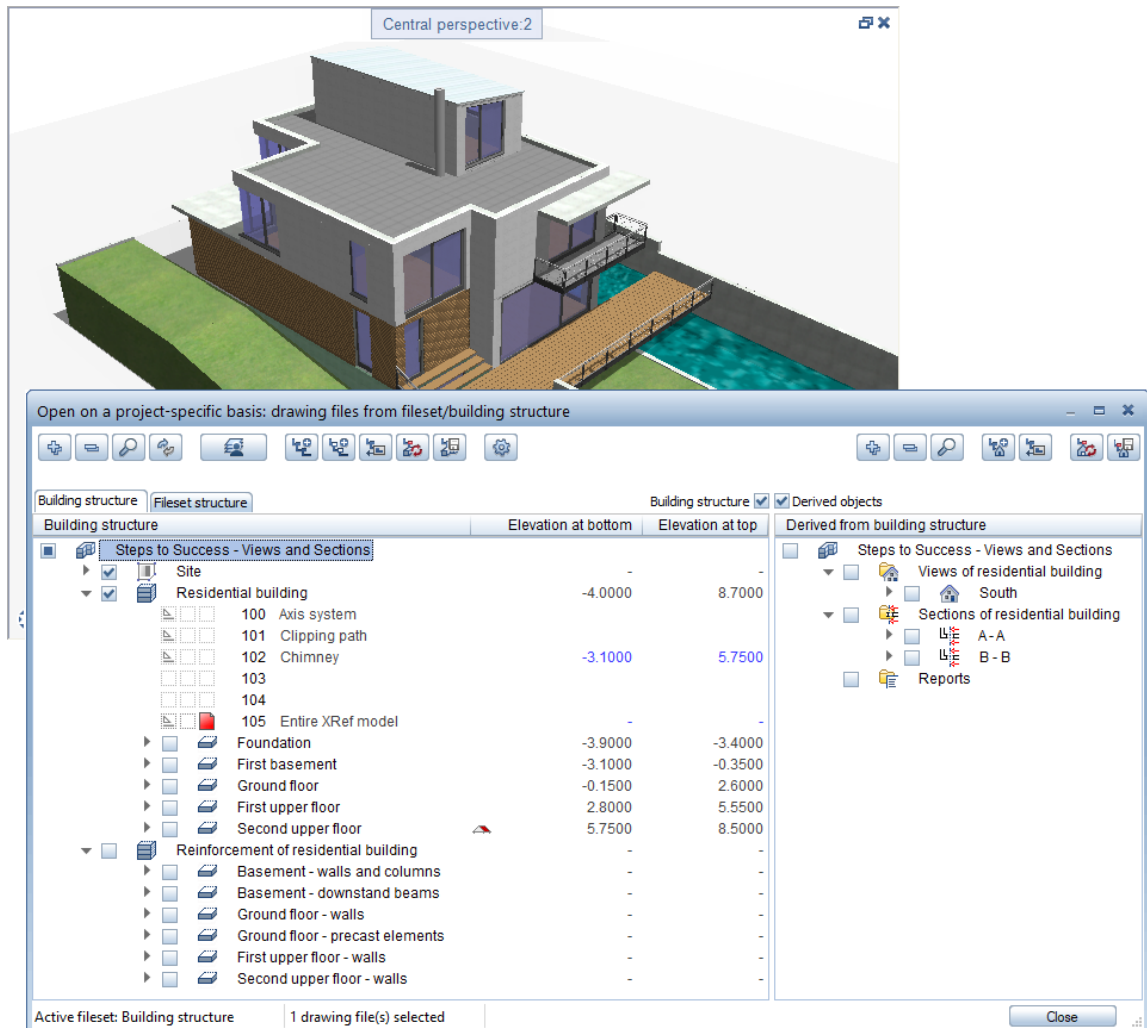
- 5 Click **OK** to confirm the **New Project, Open Project** dialog box and to open the project.
 - 6 Click  **Show/Hide** ( **View** drop-down list on the Quick Access Toolbar) and select the **Color stands for pen** option.
-

Step 1: building model

Everything you need to consider when you create the building model yourself is included in the training project. If you want, you can immediately move on to step 2 (see "Step 2: general arrangement views" on page 23).

The format properties are assigned to the components in such a way that the floor plans are displayed in section view. This way, you can use the model data for the general arrangement drawing without having to create an additional section of the floor plan.

Objective



Organization in the building structure

So that all those involved in a construction project find their way around, you always work with the building structure. Double-click in the workspace to see the structure of the building model.

Structuring the model data

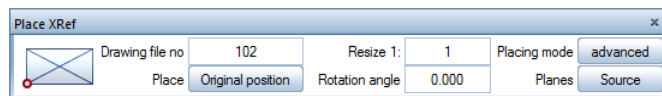
Create the structure of the model data on the left side of the **Building structure** tab. Use the following levels:

- Building
- Story
- Construction stage
- Drawing files

Adding up model data

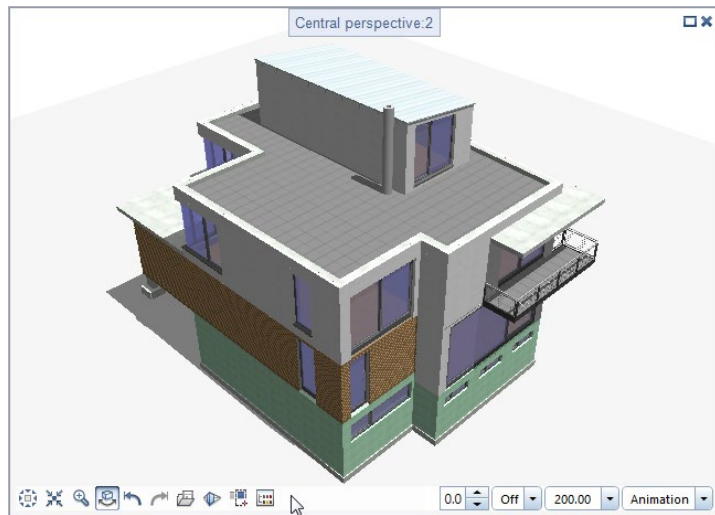
Tip: Use the Alt key to temporarily show the hidden menu bar. The menu bar disappears again after you have selected a tool.

So that you can animate and analyze the entire building model quickly and easily, place references of the required model data in a single drawing file. Use the  **XRef** tool ( **Edit** drop-down list on the Quick Access Toolbar) to place the drawing files as advanced XRefs with the following settings.



In this training project, all model data is in drawing file **105**. Select this drawing file by double-clicking it. Use **F4** to display the entire model in an additional viewport of the **Animation** view type.

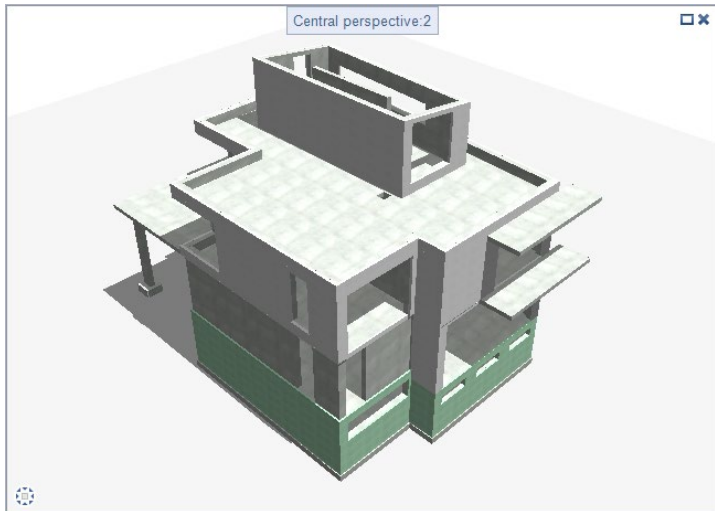
Double-click its title to maximize this viewport. To restore the previous size, double-click the title again.



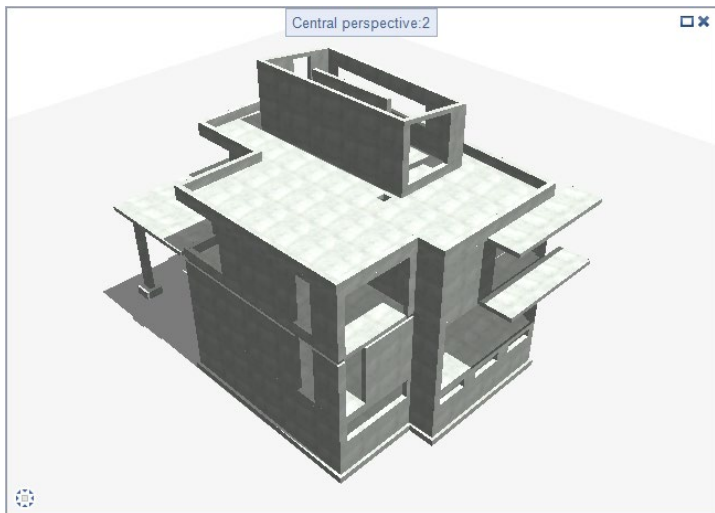
Open the  **View** drop-down list on the Quick Access Toolbar, point to **Layer Status**, click  **Select Layer Print Set**, and select the **Entire model** print set. As you can see, the finish specifications are no longer visible.



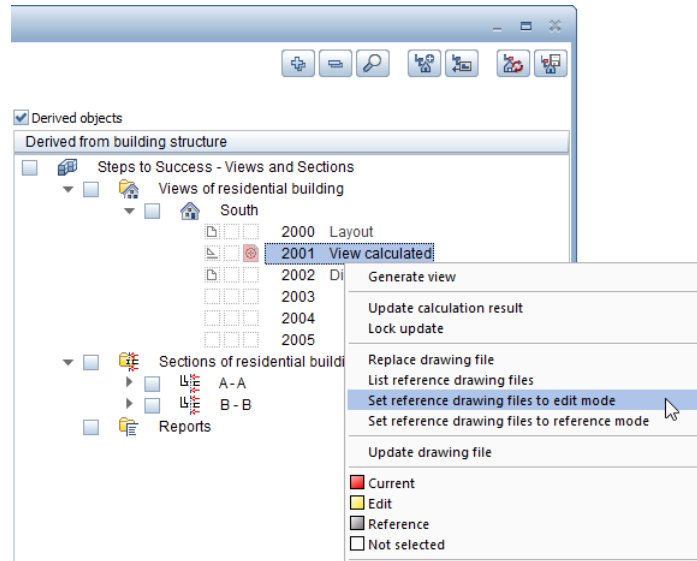
Open the  **Repeat** drop-down list on the Quick Access Toolbar, click  **Select Layer Print Set**, and select the **General arrangement drawing** print set. Only the unfinished structure with the insulation is visible.



Finally, select the **Reinforcement drawing** print set. The insulation has disappeared too.



Note: As in this training project, you can use the drawing file with the referenced model data as a source drawing file for a view or section through the entire building. To verify this, go to the "Derived from building structure" area on the right side, open the shortcut menu of the view, and select **Set reference drawing files to edit mode**.



Drawing files

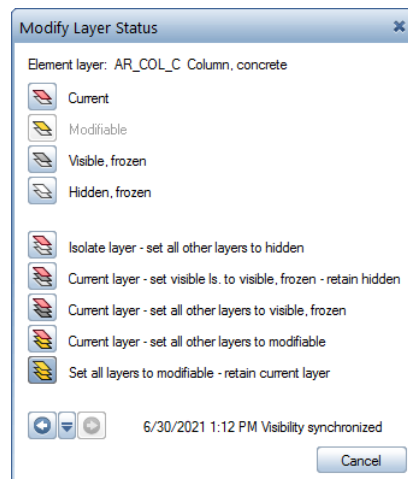
Use the same range of drawing files for the same story in all projects. Label the drawing files in such a way that you see at a glance what they contain. This is particularly important for large projects, making it easier for you to select the appropriate drawing file. For example, "1000 B1_GF_S1_walls" could stand for **walls** from construction **Stage 1** on the **Ground Floor** in **Building 1**.

Layers and print sets

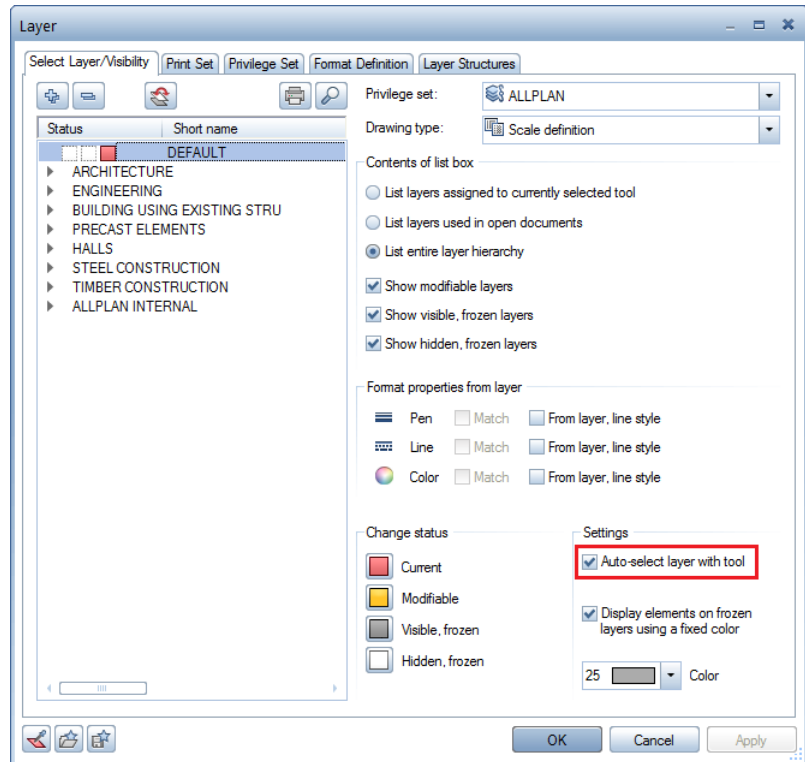
The data for the general arrangement and reinforcement is in a single drawing file. In Allplan you have various options to control what is visible without having to switch drawing files.

Layers

You can assign a layer to each element. To change the layer status, you can use the **Layer** dialog box (double-click the right mouse button in the workspace) or the **Layers** palette. You can also use the  **Modify Layer Status** tool on the shortcut menu of an element.



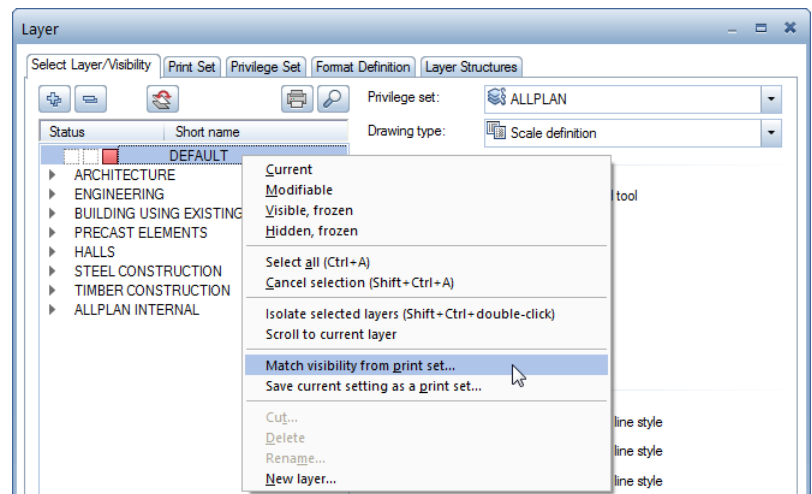
Do not change the **Auto-select layer with tool** option, which is selected by default. Otherwise, you need to reselect the layer whenever you select a tool.



Print sets

As described earlier, you can use print sets to define the visibility of all layers in a single operation.

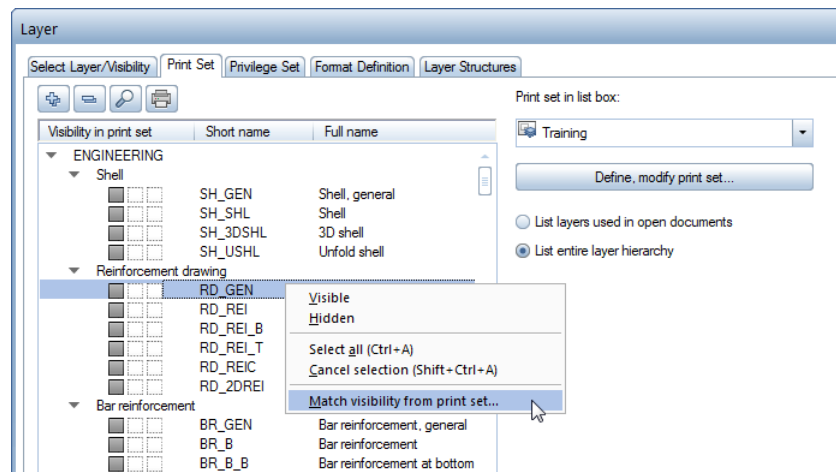
After you have defined a new print set on the **Print Set** tab, you specify which layers are visible and which hidden. You can then use these print sets to control what is visible – switch to the **Select Layer/Visibility** tab, open the shortcut menu, and choose **Match visibility from print set...**



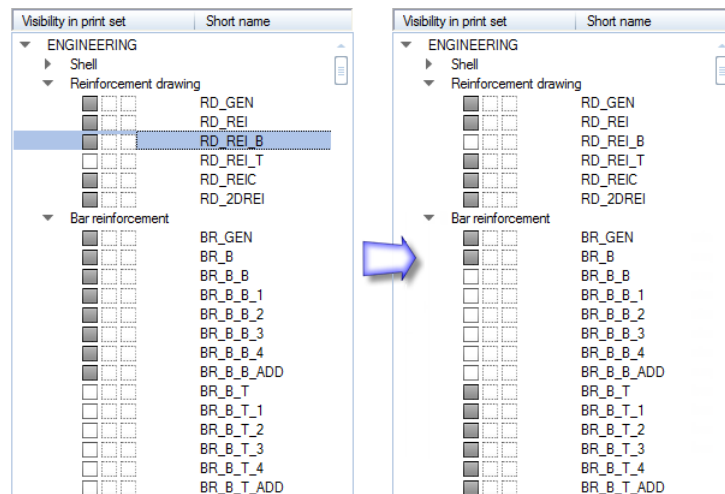
If you have already defined print sets, you can create additional print sets based on the existing ones. For example, turning the **Reinforcement, bottom layer** print set into the **Reinforcement, top layer** print set only takes a few mouse clicks.

To match the visibility of an existing print set

- 1 Click the **Define, modify print set...** button on the **Print Set** tab, create the **Training** print set, and select it in the list box.
- 2 Click **Match visibility from print set...** on the shortcut menu and select the **Reinforcement, bottom layer** print set in the dialog box.



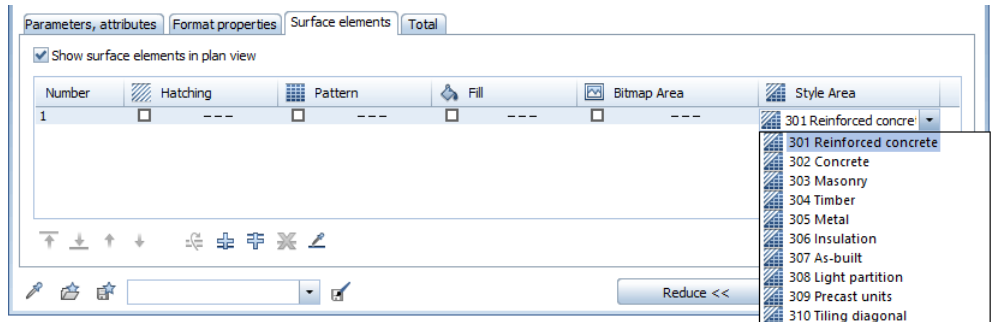
- 3 Hide the layers of the bottom layer and show the layers of the top layer for **Reinforcement drawing**, **Bar reinforcement**, **Mesh reinforcement**, and **BAMTEC**. You can do this quickly by selecting the Spacebar.



- 4 Click **OK** to confirm.

Area styles and drawing types

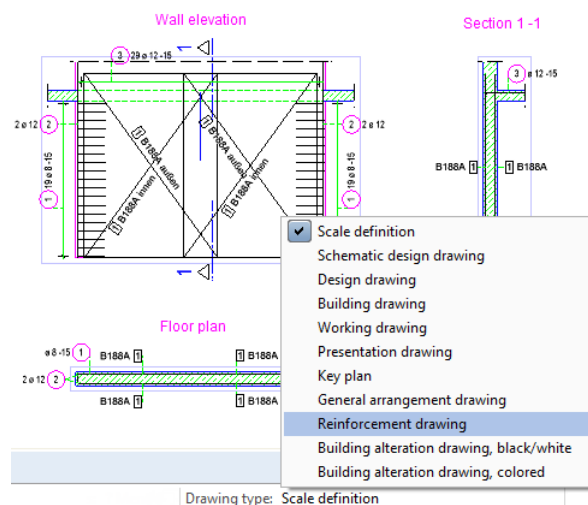
Use area styles for the components of the building model. By means of area styles, you can create different layout types without any additional drawing files or modifications.



The way area styles look changes with the drawing type.

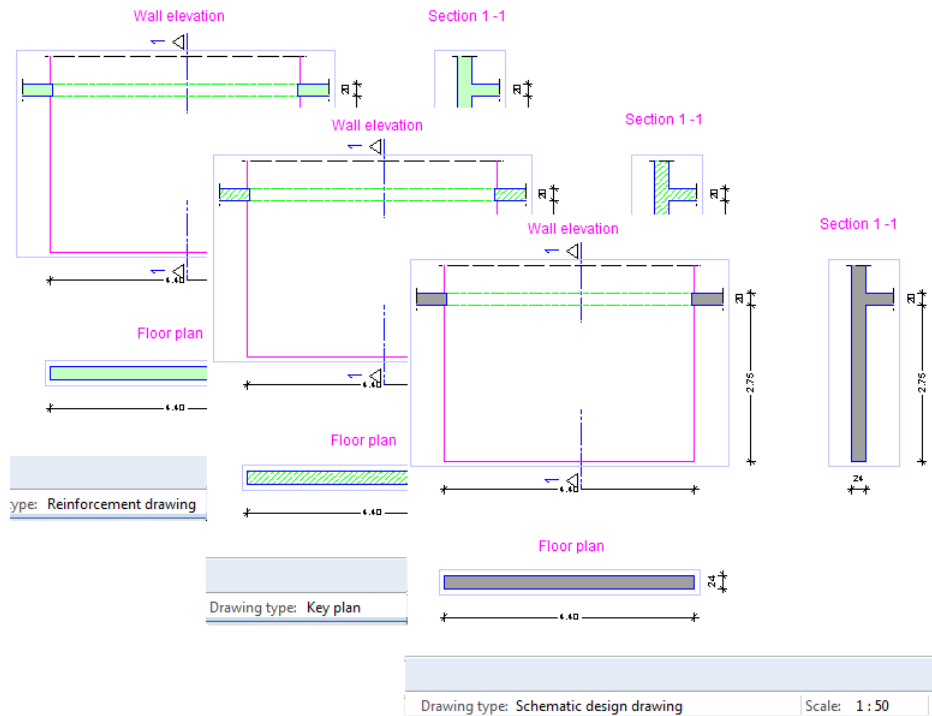
Double-click in the workspace, open the **Reinforcement of residential building – First upper floor – walls** structural levels on the left side of the building structure, and double-click drawing file 1121.

Close the viewport with the animation, zoom in on the reinforcement views, and select **Reinforcement drawing** for the drawing type on the status bar.



The on-screen display changes in accordance with the **Reinforcement drawing** print set.

Change the print set to **General arrangement drawing** and select different drawing types. This changes not only the scope of elements but also the way the elements look.



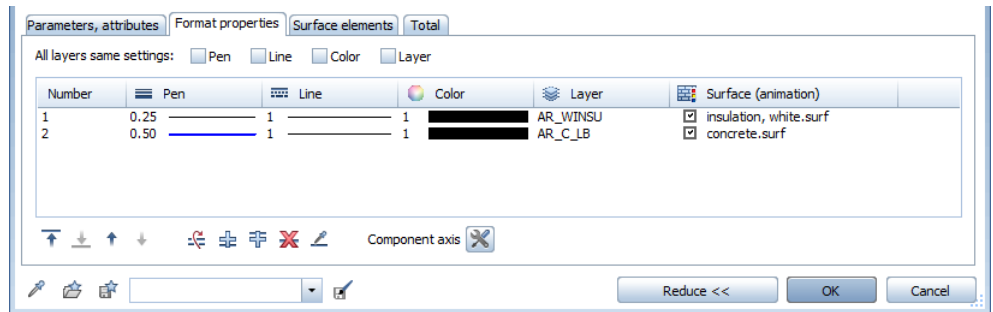
Finally, select the **Scale definition** drawing type so that the area styles change with the scale.

Experiment with different scale settings. When you have finished, change the reference scale back to 1:50.

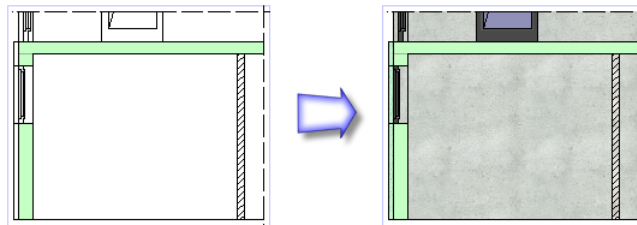
Note: You can create additional area styles and drawing types in the dialog box for layers. Open the **Format Definition** tab and click the **Manage line styles, area styles, drawing types...** button.

Textures for surfaces

Assign textures to the surfaces of the components of the building model. Textures produce realistic images for presentation purposes.



You can also use textures for the views and sections to make them look more realistic. To do this, select the  **Create bitmaps from textures, fills from colors** option in the **Surface elements** area of the **Generate View** or **Generate Section** palette. Note, however, that this requires more computing power, especially with large layouts.

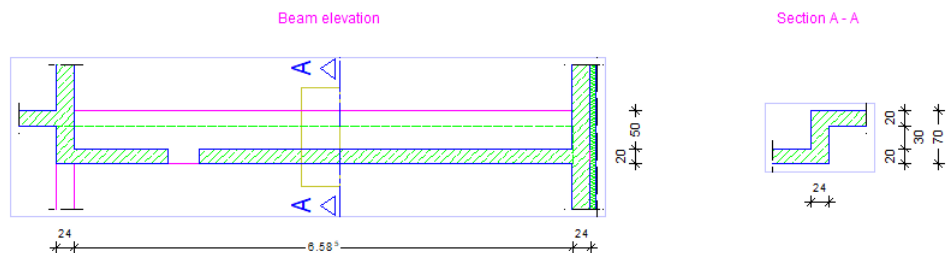


Step 2: general arrangement views

As you created the building model with the building structure, the components of the stories are in separate drawing files each having different default planes. To create the general arrangement drawings of the components, select the drawing files with the associated model data and create the required views and sections in a new, empty drawing file by using the tools in the **Sections** task area. You can find these tools on the **Actionbar**.

This approach produces links. These links generate the general arrangement views directly from the model data of the building. As you do not need to copy the data of the floor plan, the general arrangement data is always up to date.

Objective



Structuring and selecting drawing files

IFC export considers the left side of the building structure only. So that you can also transfer the reinforcement model by means of the IFC interface, create another **building** structural level with subordinate **story** structural levels to which you assign the required drawing files on the left side of the building structure.

To reduce the references, you create the general arrangement and the reinforcement in a single drawing file. So that you can work smoothly in a workgroup environment, use an additional drawing file where you place an XRef of the model data of the building for the current story.

This structure is already predefined in the training project.

Note: If you do not want to export IFC data, you can also create the required folders for the components on the right side of the building structure.

Make sure you select the appropriate drawing files when you create the views and sections across drawing files. So that you do not interrupt the workflow in a workgroup environment, load the model drawing files only when you create or modify the views and sections.

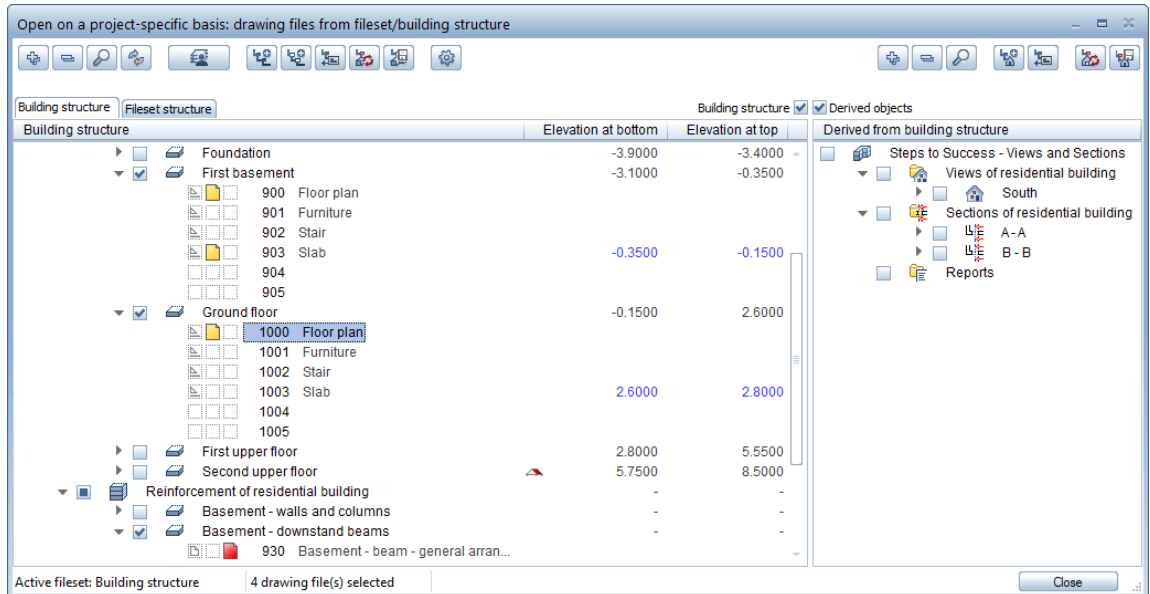
When you work across drawing files, only the views and sections are linked with the building model and reinforcement model. The model data of the building or reinforcement is not linked with the views and sections.

To make sure you select the correct drawing files when you create the views and sections, double-click the required drawing file to open only this drawing file and to make it current. Afterward, open the other drawing files in edit mode.

To provide the model data for the beam

- 1 Click  **Open on a Project-Specific Basis**, go to the left side of the **Building structure**, open the **Reinforcement of residential building – Basement – downstand beams** structural levels, and double-click drawing file **930**.
- 2 Double-click in the workspace to open the **Open on a project-specific basis** dialog box again.

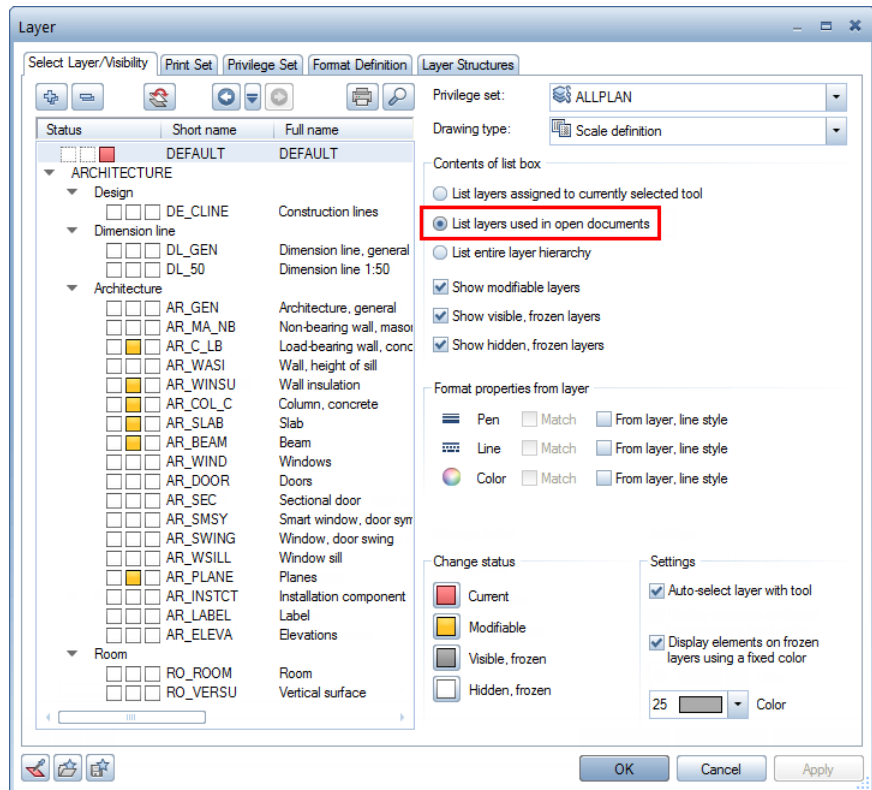
- 3 Open the **First basement** and **Ground floor** structural levels and change the status of the drawing files **900**, **903**, and **1000** to  **edit mode**. This automatically selects the structural levels.



Note: This approach ensures that only the drawing files **900**, **903**, and **1000** are selected for the model data of the beam.

- 4 **Close** the dialog box. Display everything by double-clicking the middle mouse button in the workspace. Change the scale on the status bar to **1:50**.
- 5 Open the dialog box for layers, select the **List layers used in open documents** option, and change all layers to  **Hidden**, frozen except **AR_C_LB**, **AR_WINSU**, **AR_COL_C**, **AR_SLAB**, **AR_BEAM**, and **AR_PLANE**.

Note: The layer status is correct if the **General arrangement** print set is still selected.



References and view options

Allplan creates views and sections across drawing files, placing the views and sections always in the current drawing file. If the 3D models of the elements are not in the current drawing file, Allplan creates links between the drawing files.

The following actions produce linked drawing files:

- Creating views and sections of model drawing files open in edit mode
- Creating reinforcement in loaded views and sections

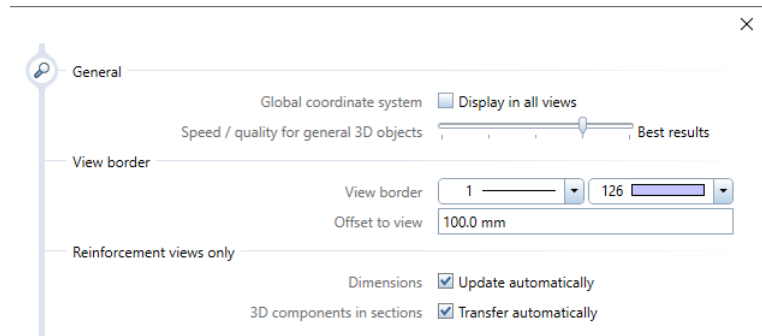
The  symbol indicates drawing files containing views and sections that you created by using the tools in the **Sections** task area. You can find these symbols in the third column of the dialog box for selecting drawing files. These views and sections might be linked with other drawing files. Allplan considers the model data in these linked drawing files.

The  symbol indicates drawing files containing views and sections that you created by using the shortcut menu in the "Derived from building structure" area or that you created by using the tools in the **Sections** task area and saved in these drawing files. You can find these symbols in the third column of the dialog box for selecting drawing files. These views and sections are usually linked with other drawing files. Allplan considers the model data in these linked drawing files.

To define view options

- 1 Click  **Options** ( **Default Settings** drop-down list on the Quick Access Toolbar) and select **Views**.
- 2 Check the settings and click **OK** to confirm the dialog box.

Note: Regardless of the setting of the **Automatically transfer 3D components to sections** option, Allplan always transfers new 3D components to views and sections that you create by using the tools in the **Sections** task area.



Creating associative sections

To reinforce the downstand beam, you need an elevation of the downstand beam and a section at midspan.

To create the front elevation

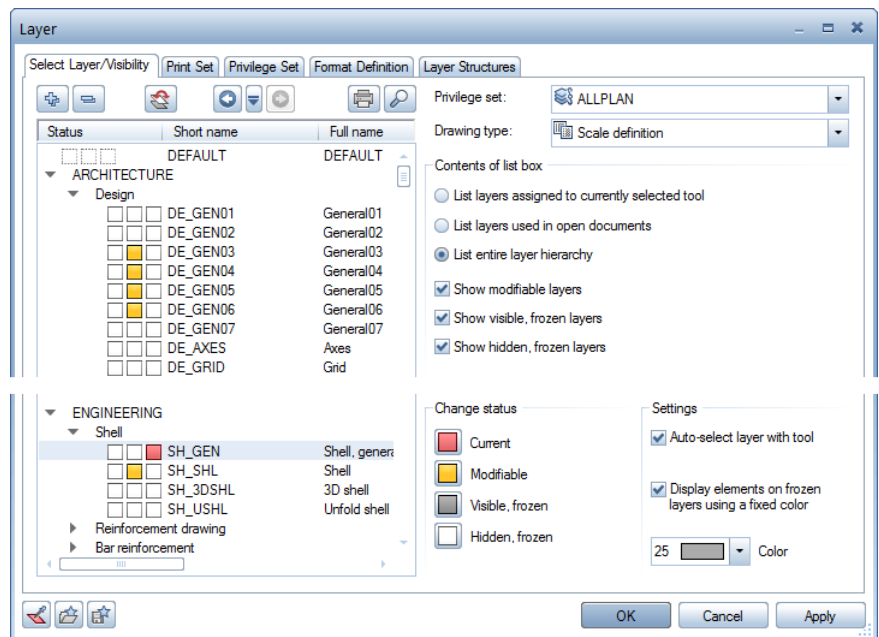
Tip: Click  to collapse the tree structure.

You can also open the **Layer** dialog box by clicking

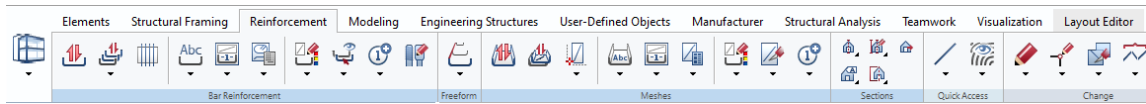
 **Expand** at the bottom of the **Layers** palette.

- 1 Open the  **View** drop-down list (Quick Access Toolbar), click  **Select, Set Layers**, and select the **List entire layer hierarchy** option. Go to **ARCHITECTURE – Design** and switch the **DE_GEN03**, **DE_GEN04**, **DE_GEN05**, and **DE_GEN06** layers to  **Modifiable**. Go to **ENGINEERING – Shell** and change the **SH_GEN** layer to  **Current** and the **SH_SHL** layer to  **Modifiable**.

Note: If the **General arrangement** print set is still selected, all you need to do is make the **SH_GEN** layer  **Current**.

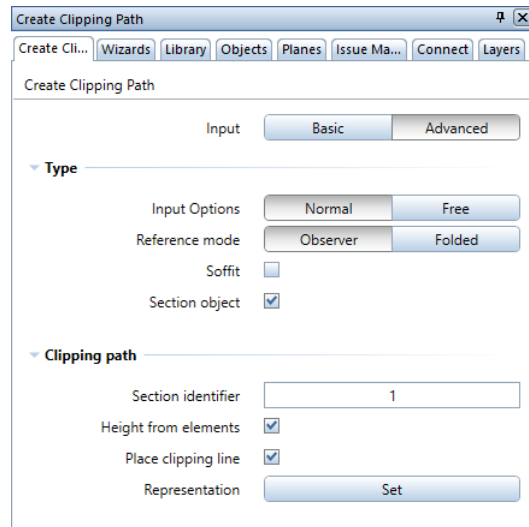


- Go to the **Actionbar**, select the  **Engineering** role, open the tab of the **Reinforcement** task, expand the **Sections** task area, and click  **Generate Section**.



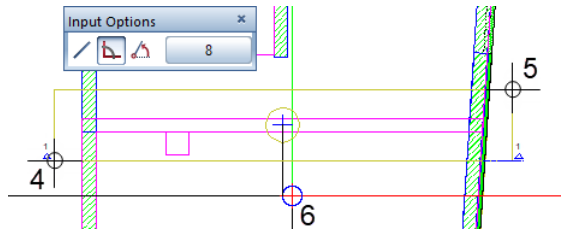
The **Create Clipping Path** palette opens so that you can define the clipping path.

- Select the **Advanced** method for **Input**. Do not change the other settings.



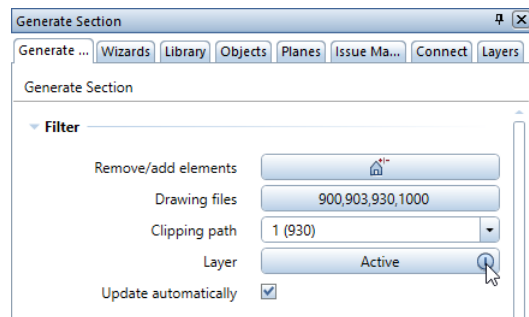
- Select view or section or enter first point:* Click to the left of and below the horizontal part of the lowered slab (see illustration).
- To point:* Click to the right of and above the horizontal part of the lowered slab (see illustration). Select the Esc key to finish entering the polyline.

- 6 *Select viewing direction:* Define the viewing direction by clicking below the circle. As a result, Allplan views the object from the front when calculating the section.



The **Generate Section** palette opens; the section is attached to the crosshairs. The label uses the layer in the **Properties** palette – **Format** area. The view border is always on the DEFAULT layer, regardless of this setting. The section object uses the layer that you select for the **Clipping lines** in the **Create Clipping Path** palette. The layers for the elements of the section come from the 3D components, or you can specify the layers when you define the formats for edges, clipping lines, and finish elements on the tabs.

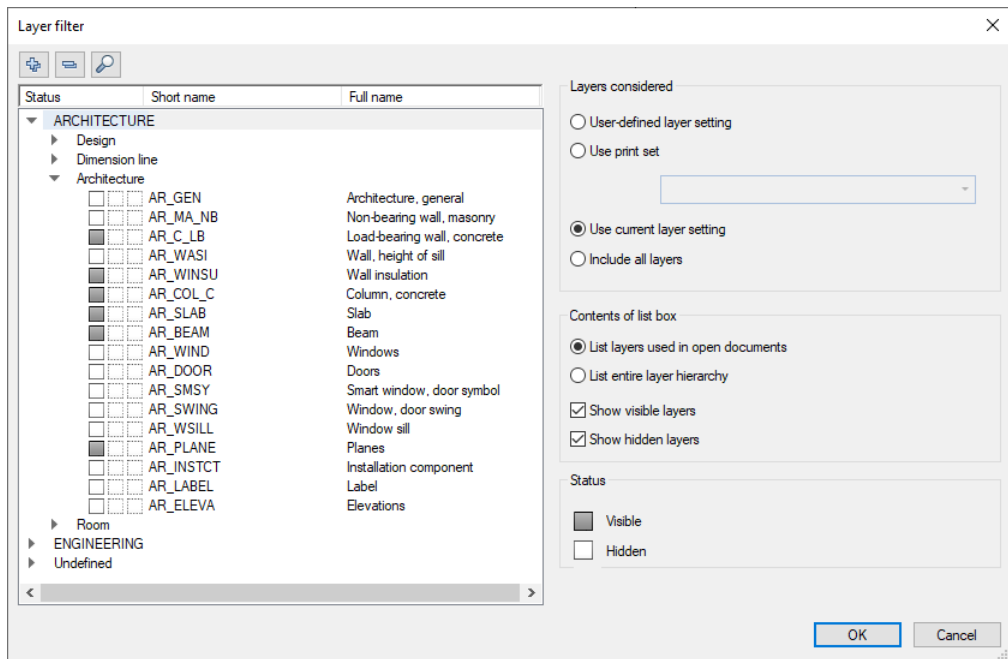
- 7 Go to the **Filter** area of the palette and turn the **Layer**  **Off**.



The text on the button changes from **Active** to **Set**, which means that the section uses the current layer setting.

- 8 Click the **Set** button to open the **Layer filter** dialog box. Select the **List layers used in open documents** option and click **OK** to confirm the dialog box.

Note: If you want to permanently use the elements of the current layer setting for the section, select the **User-defined layer setting** option. In this case, switch the layer of the reinforcing bars to  **Visible** in the layer filter of the views and sections before you create the reinforcement.

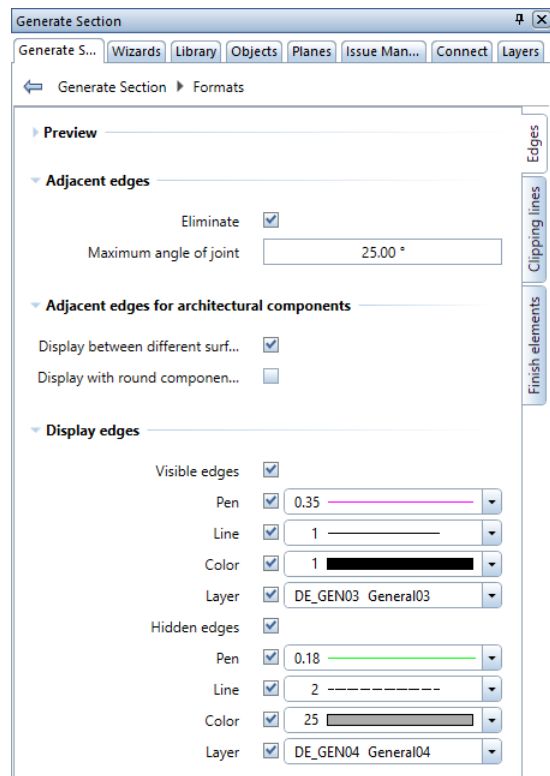


- 9 Do not change the other settings. Go to the **Representation** area and click the **Set** button to the right of **Formats**.

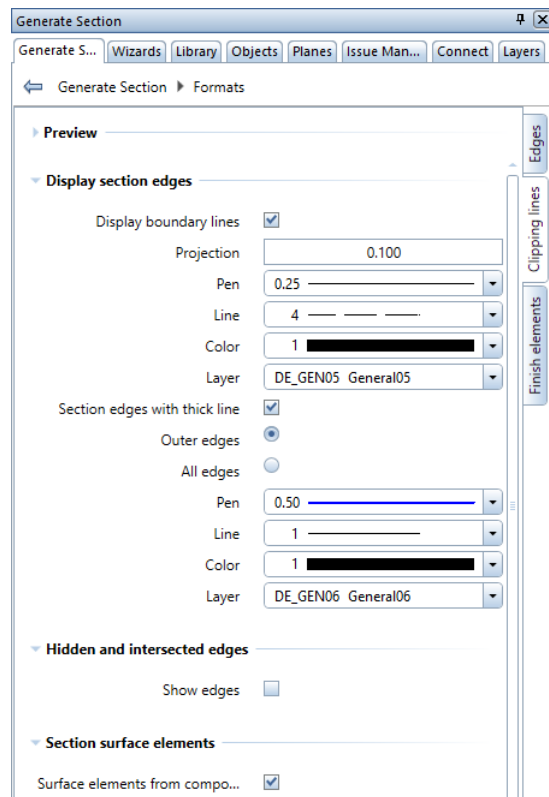
The **Formats** palette opens; you can see the **Edges** tab.

10 Check the **Display edges** area to make sure that the **Visible edges** and **Hidden edges** options are selected. Define the following format properties.

- Visible edges:
Select pen **0.35** mm; do not change the line or color; select layer **DE_GEN03**
- Hidden edges:
Do not change the pen, line, or color; select layer **DE_GEN04**

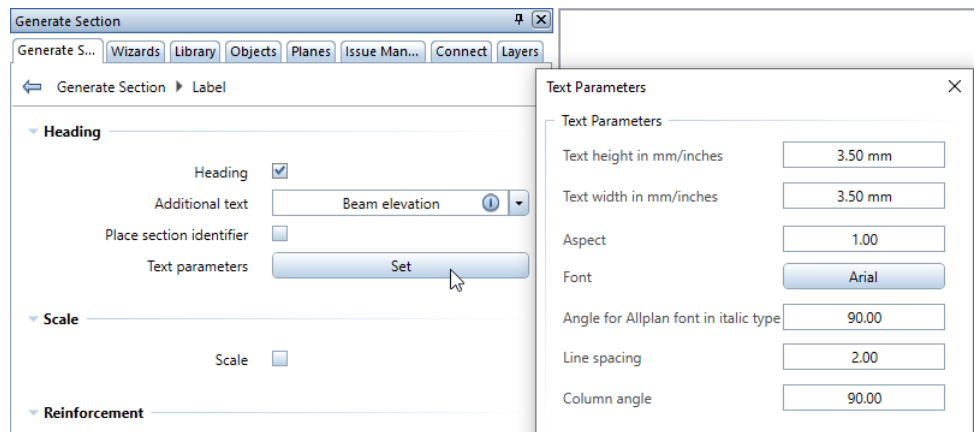


- 11 Switch to the **Clipping lines** tab and define the following settings:
 - Select the **DE_GEN05** layer for the boundary lines.
 - Select the **Display section edges with a thick line** option in the **Display section edges** area. Select the **Outer edges** option and the **DE_GEN06** layer.
 - Do not change the other settings.



- 12 Switch to the **Finish elements** tab and clear the **Show floor surfaces** check box.
- 13 **Close** the palette to save the settings and to return to the **Generate Section** palette.

- 14 Check the **Representation** area to make sure that the **Show clipping path** option is selected. Click the **Set** button to the right of **Label**.
- 15 The **Label** palette opens. Enter the label for the section in the **Additional text** box:
Beam elevation
Clear the **Place section identifier** check box.
- 16 Click the **Set** button to the right of **Text parameters**, define the parameters for the label (text height and width of **3.5 mm**), and click **OK** to confirm the dialog box.



- 17 *To point or angle of rotation:* Place the section to the right of the floor plan.

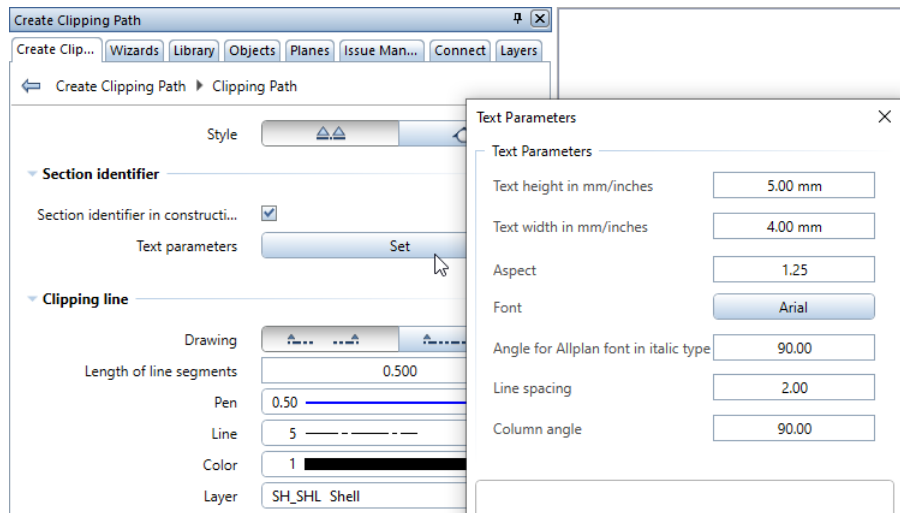
The next step is to create the cross-section based on the section you just created.

To create the required section

- The **Create Clipping Path** palette is still open. If it isn't, open this palette by selecting the  **Generate Section** tool again.

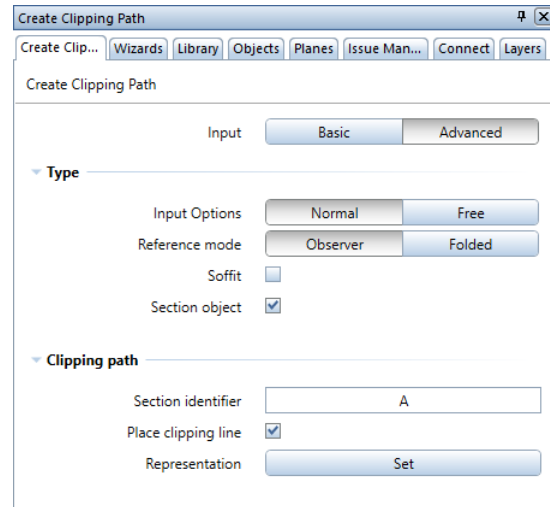
Tip: The **Observer** reference mode creates the section so that horizontal component edges are parallel to the global x-axis. In other words, horizontal edges are always horizontal, regardless of the viewing direction. In contrast, the **folded** reference mode simply folds the section.

- 1 In the **Type** area, you can switch between the **Observer** and **Folded** reference modes. **Observer** is selected; do not change this setting.
- 2 Go to the **Clipping path** area and change the **Section identifier** to **A**. Click the **Set** button to the right of **Representation**.
- 3 The **Clipping Path** palette opens. Go to the **Section identifier** area and click the **Set** button to the right of **Text parameters**. Enter **5.0 mm** for the text height and **4.0 mm** for the text width. Then, click **OK** to confirm the dialog box.



- 4 Select the **AR_SECT** layer for the **Clipping line**. Click **Close** at the bottom of the palette to save the settings and to return to the **Create Clipping Path** palette.
- 5 *Select view or section or enter first point:* Click the view border of the section you just created to define the clipping path within the beam elevation.

The **Height from elements** option is no longer visible. The height that is perpendicular to the workspace comes from the beam elevation clicked. As the viewing direction of the beam elevation is not in the z-direction, the top level and bottom level of the new section are defined by the clipping path that you enter.



- 6 *First point of clipping path – second point:* Click two diagonal points at midspan in the beam elevation. Select ESC to finish entering the polyline.
- 7 *Select viewing direction:* Define the viewing direction by clicking to the right of the circle. As a result, Allplan views the 3D elements from the right when calculating the section.
- 8 Go to the **Representation** area and turn off the **Show clipping path** option. Click the **Set** button to the right of **Label**.
- 9 Select **Section** in the **Additional text** list box and select the **Place section identifier** option.

Tip: Track tracing helps you place points in exact alignment with existing points. You can select the **F11** key or click  in the dialog line to turn track tracing on and off.

- 10 *To point or angle of rotation:* Place the section so that it is to the right of and aligned with the beam elevation.

The **Create Clipping Path** palette opens again. Allplan creates any other clipping paths within the selected beam elevation until you finish creating sections within a view or section by selecting ESC.

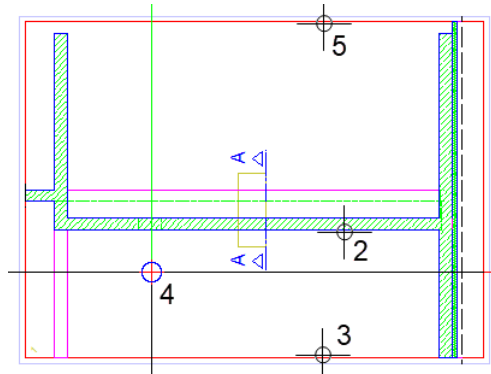
- 11 Select ESC twice to close the tool.
-

The next step is to reinforce the beam. For this purpose, you display only the connections to the adjacent components.
The right support area at the inclined exterior wall is visible over the full depth of the section. This is not useful for reinforcing the beam. Therefore, adjust the clipping area of the beam elevation in the model.

To adjust the front elevation and to add dimensions

- 1 Right-click in the workspace and select  **Modify Offset** on the shortcut menu.
- 2 *Click reference line:* Click a horizontal line (for example, the bottom of the lowered slab) in the beam elevation.
- 3 *Click line:* Click the bottom line delimiting the section object.
The dialog line shows the distance between the lines clicked.
- 4 *Through point or offset:* Click to define the point through which the bottom line delimiting the section object is to pass or enter the required value in the dialog line.

- 5 *Click line:* Click the top line delimiting the section object and modify the distance in the same way.



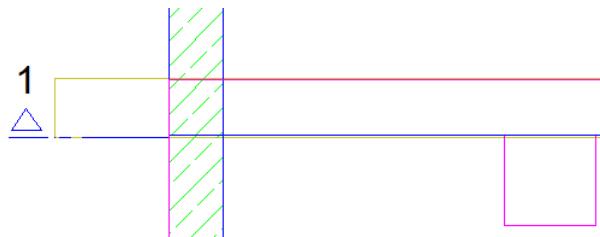
Tip: You can enter the values for the **Top level** and **Bottom level** of the section when you define the clipping path. To do this, clear the **Height from elements** check box.

- 6 Select ESC to close the tool.
7 Click  **Repeat** on the Quick Access Toolbar.

This opens the last tool you selected. In this example, it is  **Modify Offset**.

- 8 Modify the section object in the floor plan so that the section ends just beside the beam.

For example, enter **0.01** in the dialog line so that the section does not end with the beam edges. As a result, the adjacent slabs are not visible.



Tip: You can use  **Link Dimension Lines to Views** to link nonassociative dimension lines to views and sections later.

- 9 Use  **Dimension Line** (Quick Access task area) to add the dimensions of the beam to the beam elevation and the section. Make sure that you create associative dimension lines. Open the  **Properties** and select the **SH_SHL** layer for the dimension-line elements.

Tip: To hide the section objects in the floor plan, you can use the  **Modify Clipping Path** tool.

- 10 Right-click the view border of the beam elevation and select  **Show Clipping Path** on the shortcut menu.

- 11 *Select section object:* Click the modified section object in the beam elevation or the floor plan.

Note: If you open the tool by using the shortcut menu of the section object, you must click the section.

If you have inadvertently hidden the clipping path of section A-A in the beam elevation or if you want to show the section object of the beam elevation again, click the required clipping path in the floor plan and then click the beam elevation.

- 12 Select ESC to close the tool. Make the **DEFAULT** layer current again.

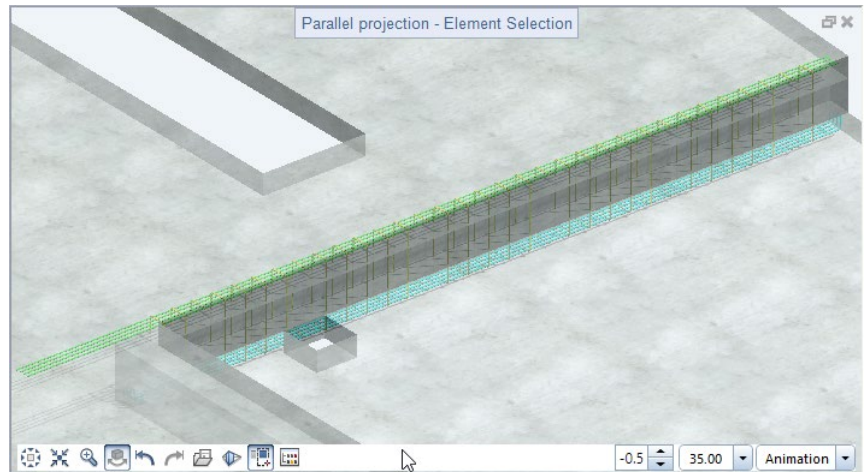
- 13  **Move** (shortcut menu or **Edit** task area) the labels of the sections to make space for the reinforcement labels that you will create later.
-

Step 3: reinforcement

After turning off the model data, you create part of the reinforcement in the general arrangement views of the downstand beam. So that you can use the drawing file with the associated sections for both the general arrangement drawing and the reinforcement drawing, it is important that you place the reinforcement on layers.

If you create the general arrangement and reinforcement in different drawing files, the reinforcement model is in the current drawing file whereas the reinforcement is always in the drawing file with the views and sections.

Objective



Note: The **Concrete_reinforced.surf** surface (transparency: 50%) has been assigned to the concrete components so that the reinforcement is displayed in the animation of the reinforcement drawing. Alternatively, change the transparency setting of the **Concrete.surf** surface and save it as a new surface. However, this change applies to all the concrete components in the entire project.

Creating reinforcement with a 3D model

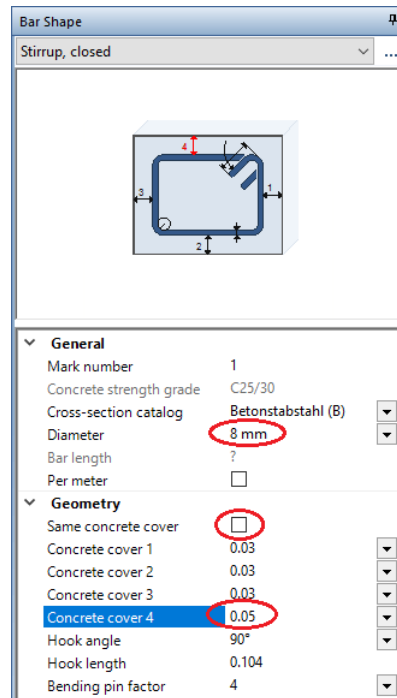
Create the stirrup reinforcement and the longitudinal bars at the bottom and top as part of the beam reinforcement.

This section does not show how to create the reinforcement for the wall restraints, the starter bars for the slab, the web reinforcement, or the reinforcement schemas because this would go beyond the scope of this guide.

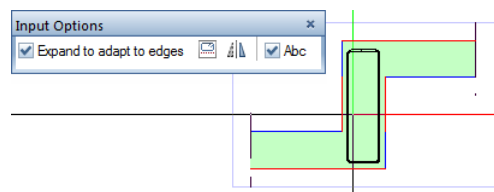
To create expanding stirrup reinforcement

- The **Reinforce with 3D model** option is selected in the  **Options** on the **Reinforcement** page.
- **Actionbar:**  **Engineering** role – **Reinforcement** task.
- 1 Open the **Layers** palette, click  **Select Layer Print Set** in the lower-right area of the palette, and select the **Reinforcement drawing** print set. Select **Reinforcement drawing** for the drawing type on the status bar.
- 2 Double-click in the workspace. Go to the **Basement – downstand beams** structural level and double-click drawing file **930**.
- 3 Zoom in on the beam elevation and the section. Check that the scale on the status bar is **1:50**.
- 4 Go to the **Actionbar – Bar Reinforcement** task area and click  **Bar Shape**. Open the **Layers** palette and select the **BR_B** layer.
- 5 Switch to the **Bar Shape** palette and select the **Stirrup, closed** bending shape in the list box at the top.

- 6 Go to the area with the parameters in the palette, switch the diameter to **8 mm**, clear the **Same concrete covers** check box, and change **Concrete cover 4** to 0.05.



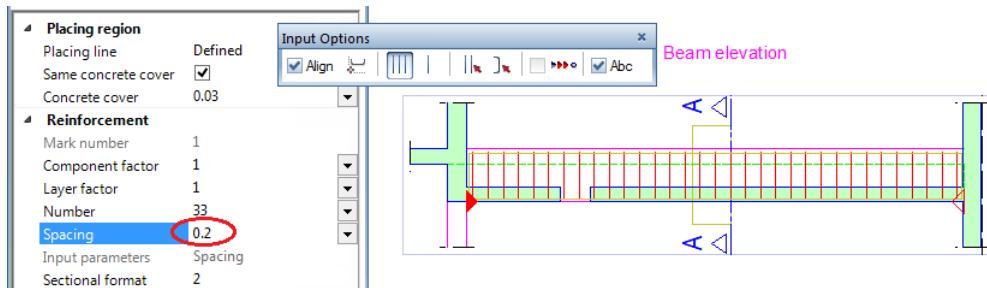
- 7 The **Expand to adapt to edges** and **Label** options are selected in the input options. Go to section A-A, point to the left component line, wait until the bending shape expands, and click.



- 8 Use ESC to select the  **Label** tool. Place the bar label with the mark number and diameter to the right of the bar.

The  **Place Bar Shape** tool starts automatically.

- 9 Define the clear opening of the beam as the placing region by clicking the points at the lower side of the beam in the elevation. Select **Align** in the input options. The **Label** option is selected; do not change this.
- 10 Enter **0.20** for the spacing and use ESC twice to close the tool and to select the  **Dimension Line, Label** tool.

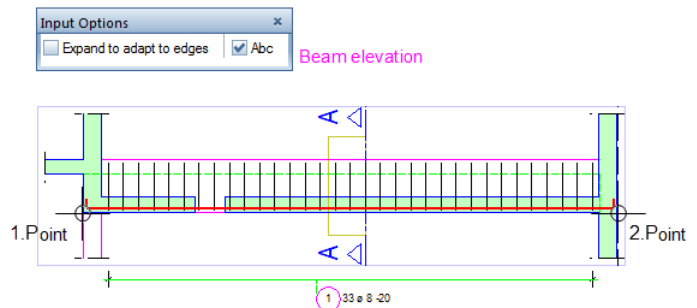
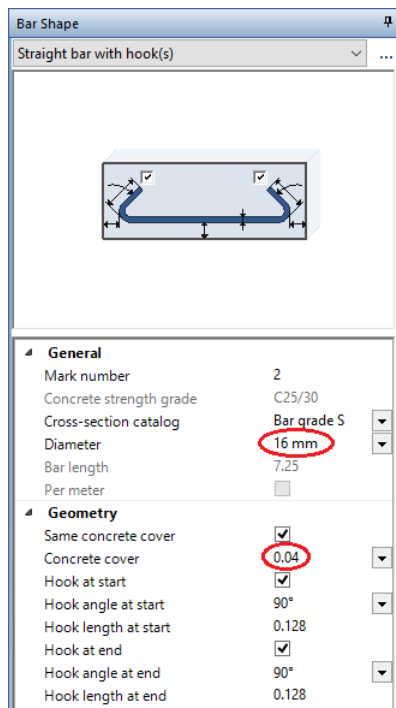


- 11 Select the **Dimension line** type in the list box at the top. Place the dimension line and the label below the placement. Define the label parameters; include the number of pieces and the diameter.
- 12 Select ESC to finish creating the stirrup reinforcement.

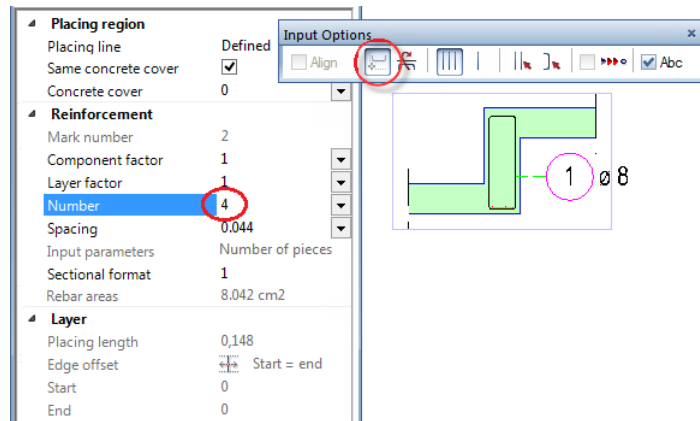
Place 4 bars of diameter 16 with end hooks for the longitudinal reinforcement at the bottom. The longitudinal reinforcement at the top consists of 4 bars of diameter 12 that extend into the slab in the area of the interior column.

To create longitudinal reinforcement at the bottom and top

- 1 Open the  **Repeat** drop-down list, click  **Bar Shape**, and select the **Straight bar with hooks** bending shape in the list box at the top of the **Bar Shape** palette.
- 2 Go to the area with the parameters in the palette, switch the diameter to **16 mm**, and change the **Concrete cover** to **0.04**.
- 3 Clear the **Expand to adapt to edges** check box in the input options, click the end points at the bottom of the beam in the elevation, and select ESC.

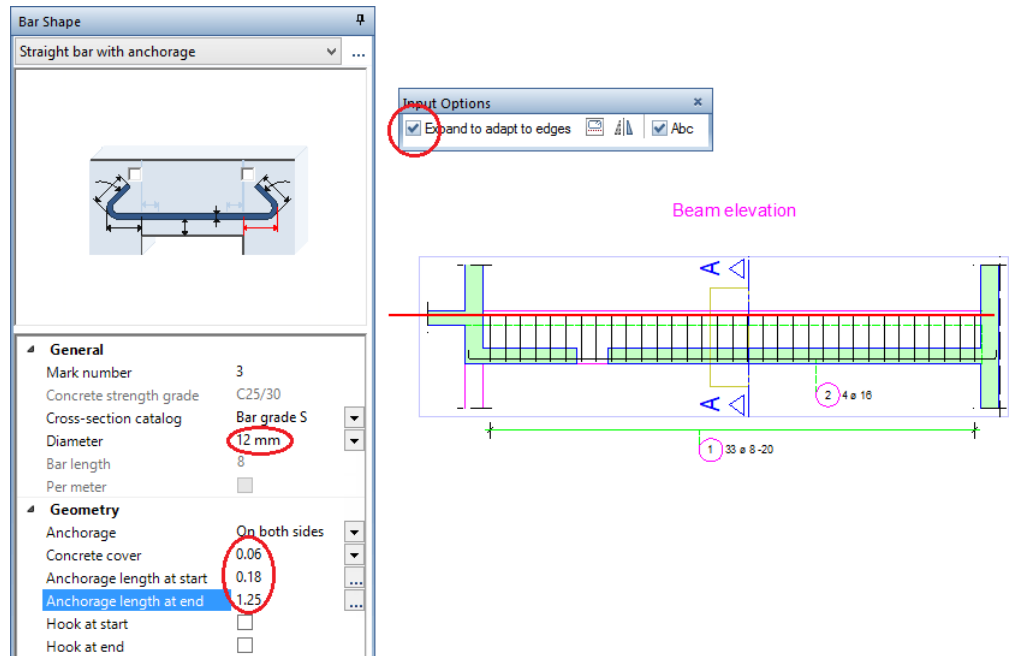


- 4 Place the label with the mark number, number of pieces, and diameter.
- 5 Click  **Segment** in the input options and click the stirrup segment at the bottom in section A-A.
- 6 Enter **4** for the number of pieces and select ESC twice.



- 7 Select the **Fan** dimension-line type and place the label with the mark number, number of pieces, and diameter.
- The  **Bar Shape** tool is still open.
- 8 Select the **Straight bar with anchorage** bending shape in the **Bar Shape** palette.
 - 9 Go to the area with the parameters in the palette, switch the diameter to **12 mm**, change the **Concrete cover** to **0.06**, and clear the **Hook at start** and **Hook at end** check boxes.
 - 10 Select **Expand to adapt to edges** in the input options, go to the elevation, point to the component line at the top, wait until the bending shape expands, and click.

- 11 Enter **0.18** for the **Anchorage length at start** and **1.25** for the **Anchorage length at end**.



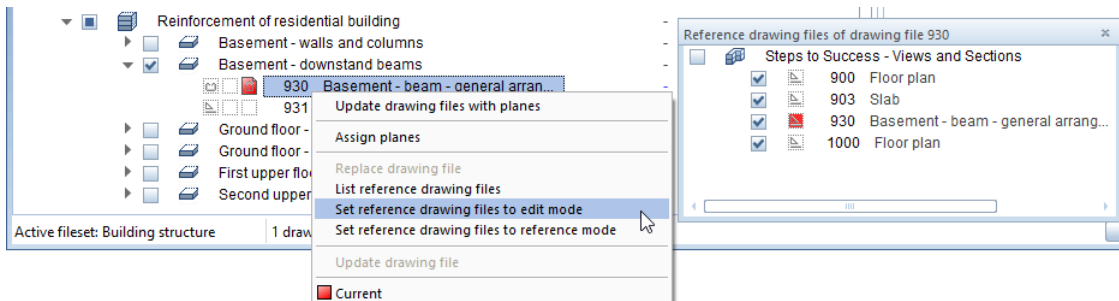
- 12 Label and place the longitudinal reinforcement at the top as described for the longitudinal reinforcement at the bottom.
- 13 Select ESC to finish creating the longitudinal reinforcement.

Selecting drawing files for modifying data

To complement and modify the general arrangement and reinforcement, you use the option to set the reference drawing files to edit mode.

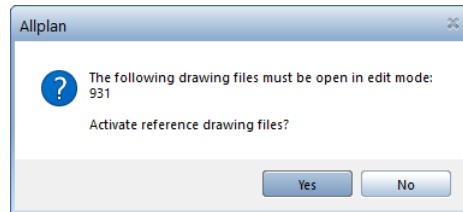
To select the drawing files for modifying data

- 1 To change the general arrangement in the existing views and sections or to add more views and sections, open the dialog box for selecting drawing files and right-click the drawing file with the associative views and sections, that is, drawing file **930** in this example.
- 2 Select **List reference drawing files** to check the existing links.
- 3 If everything is okay, open the shortcut menu of drawing file **930** again, click **Set reference drawing files to edit mode**, and **Close** the dialog box for selecting drawing files.



- 4 To change the reinforcement in the existing views and sections or to enter more reinforcement, you need to open only the drawing file with the reinforcement model in edit mode. In this example, only drawing file **930** must be open in edit mode. But if you create the general arrangement and reinforcement in different drawing files, double-click the drawing file with the reinforcement model and then open the drawing file with the views and sections in **edit mode**.

Note: If you want to modify reinforcement in views and sections and the model data of the reinforcement is not open in edit mode, a message tells you which drawing files need to be open in edit mode. By confirming this message, you can directly open the required drawing files in edit mode without having to open the dialog box for selecting drawing files.



Using reinforcement again

A construction project often contains the same components with identical or at least similar reinforcement for which you must create a separate drawing. In this training project, each floor has the same wall next to the interior stair.

The drawing file **920** of the **Basement – walls and columns** structural level contains the associative sections and the reinforcement model for the wall next to the interior stair in the first basement. So that you do not need to re-create the general arrangement and reinforcement for the ground floor, copy the drawing file in the building structure, move the contents of the drawing file, and adjust the linked model data afterward.

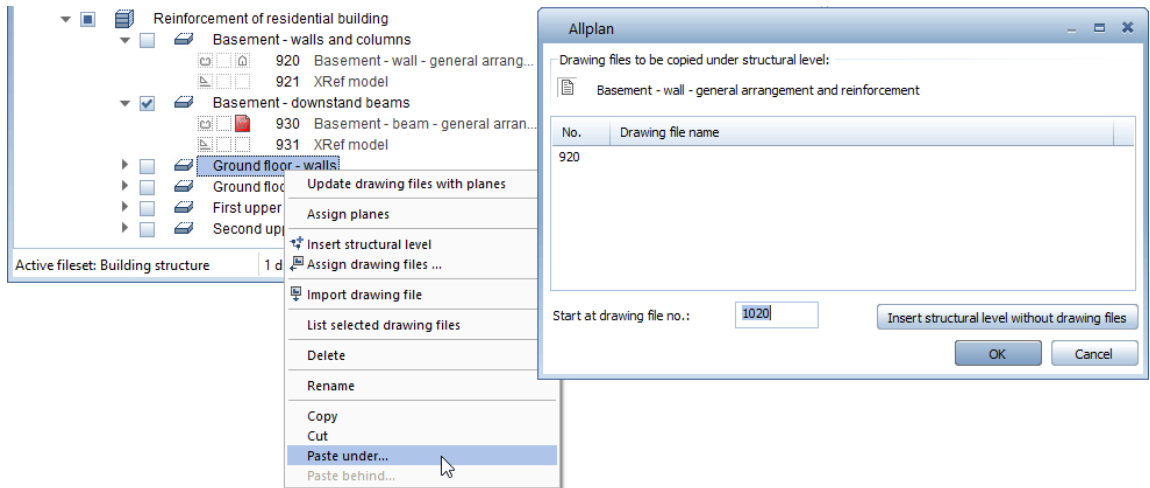
To copy the general arrangement and reinforcement

- 1 Double-click in the workspace and open the **Basement – walls and columns** structural level on the left side of the building structure. Open the shortcut menu of drawing file **920** and select **Copy**.

Note: If you created the general arrangement and reinforcement in different drawing files, you need to select both drawing files to **copy** them. When you paste the drawing files afterward, Allplan re-creates the contents of the drawing files in different drawing files; the views and sections copied get links to themselves and to the reinforcement model copied.

- 2 Open the shortcut menu of the **Ground floor – walls** structural level and select **Paste under**.

- 3 A dialog box opens. Enter **1020** for **Start at drawing file no.** and click **OK** to confirm.



This copies the contents of drawing file 920 including the drawing file name to drawing file 1020. Allplan adjusts the link.

- 4 Rename drawing file 1020 **Ground floor – wall – general arrangement and reinforcement** and double-click drawing file 1020.

To adjust the copied data

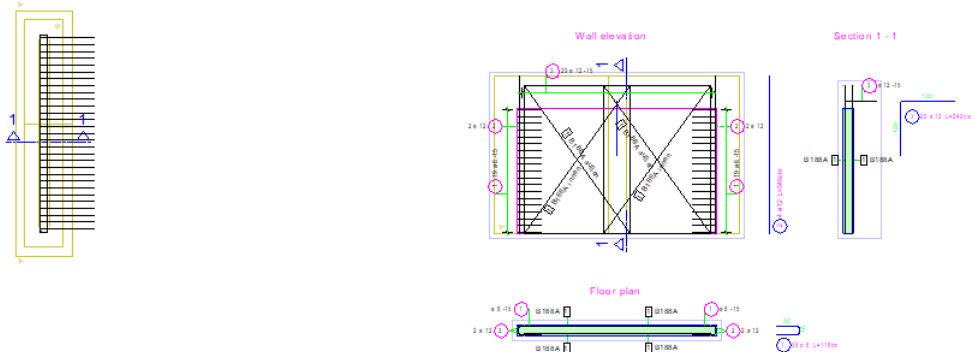
➔ Actionbar:  Engineering role – Reinforcement task.

- 1 Go to the **Actionbar – Sections** task area and click  **Move Linked**. Use CTRL+A to select everything.

Note: If you created the general arrangement and reinforcement in different drawing files, you need to select both drawing files to **move** them. Otherwise, the representation and label of the placement will be lost.

- 2 As the wall is congruent on the next floor, move everything by 2.95 m (= story height) in the z-direction.

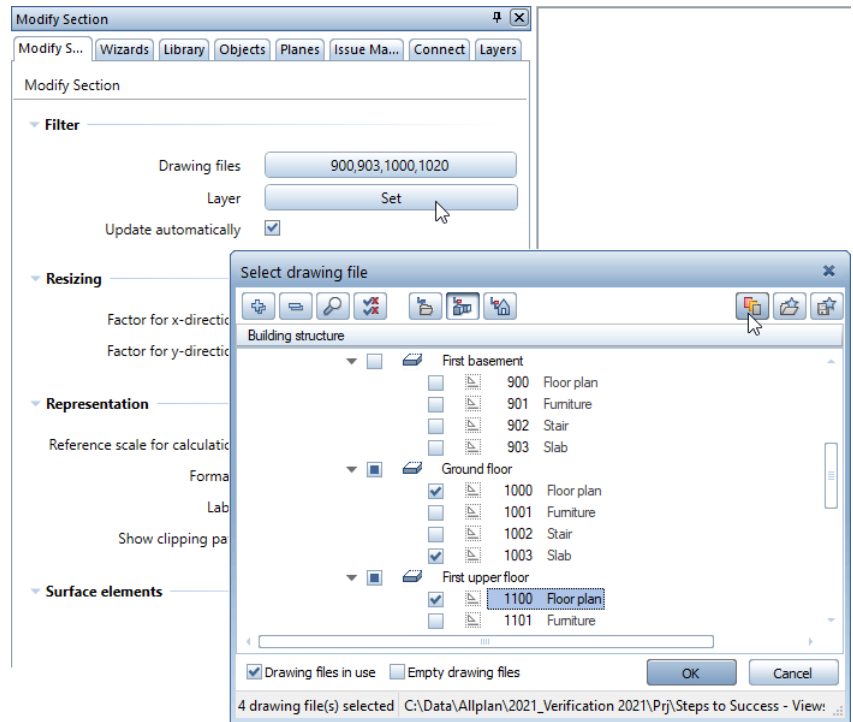
The representation of the general arrangement changes in the wall elevation and section because the views and sections are still linked with the original model data, but the reinforcement is now on the next floor. The representation of the reinforcement and the labels remain unchanged.



- 3 Go to the **Actionbar – Sections** task area and click  **Modify View, Section**. Use CTRL+A again to select all sections. You can also enclose the sections in a selection rectangle.
- 4 Click the button to the right of **Drawing files** in the **Filter** area of the **Modify Section** palette.
- 5 You can see the **Select drawing file** dialog box. Click  **Building structure** and then  **Select drawing files currently loaded** on the toolbar at the top.

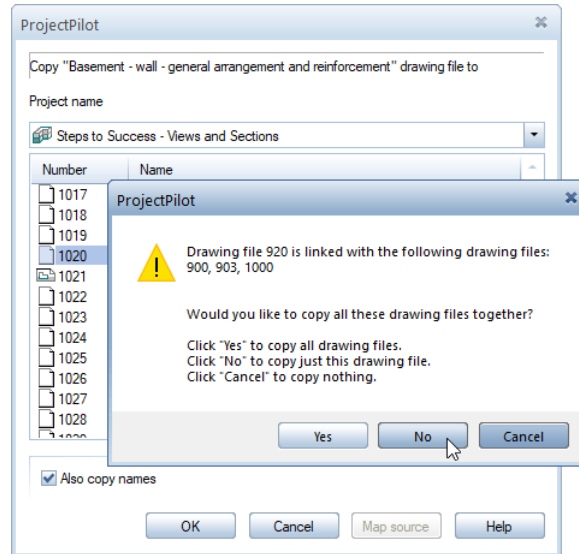
Only drawing file **1020** is selected in the building structure. Therefore, the drawing files **900**, **903**, and **1000** will be closed automatically.

- 6 Go to the **Ground floor** and **First upper floor** structural levels and select the check boxes of drawing files **1000**, **1003**, and **1100**.



- 7 Click **OK** to confirm the dialog box. This deletes the links to the architectural data from the first basement and creates links to the architectural data on the ground floor.
- 8 Click **Apply** and **Close** at the bottom of the palette to save the changes and to close the tool.

Alternatively, you can also use  **ProjectPilot** to copy the associative sections and the reinforcement model. In this case, select the required drawing files one after the other. Click **No** when ProjectPilot asks whether you want to copy all linked drawing files together.



If you created the general arrangement and reinforcement in different drawing files, you need to copy both drawing files in a single operation to create the appropriate links.

When you copy drawing files with views and sections by using the  **Copy, Move Elements between Documents...** tool or the clipboard, the link to the original drawing file remains unchanged. Consequently, you need to delete this link afterward as described in this section and in "Managing reference drawing files (on page 55)". But if you want to have the general arrangement and reinforcement model in different drawing files afterward, use this approach.

Managing reference drawing files

If other drawing files are open in **edit** or **reference mode** while you are creating views or sections and you do not close these drawing files by using the drawing file filter, the views and sections get links to these drawing files.

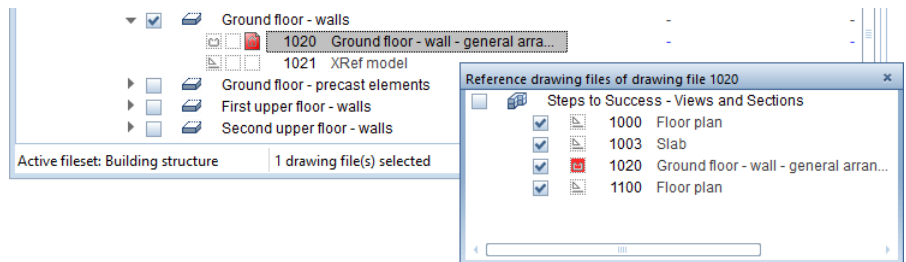
When creating views and sections, you can only link drawing files that are selected when you open the tool. When modifying views and sections, however, you can define any drawing file as a reference drawing file. If you want to link or unlink a drawing file while creating or modifying views and sections, click the button to the right of **Drawing files** in the **Filter** area of the palette and turn the required drawing file on or off in the **Select drawing file** dialog box.

As mentioned in the preceding section, the link to the original drawing file remains unchanged when you copy drawing files with views and sections by using the  **Copy, Move Elements between Documents...** tool or the clipboard. Therefore, you need to delete this link afterward.

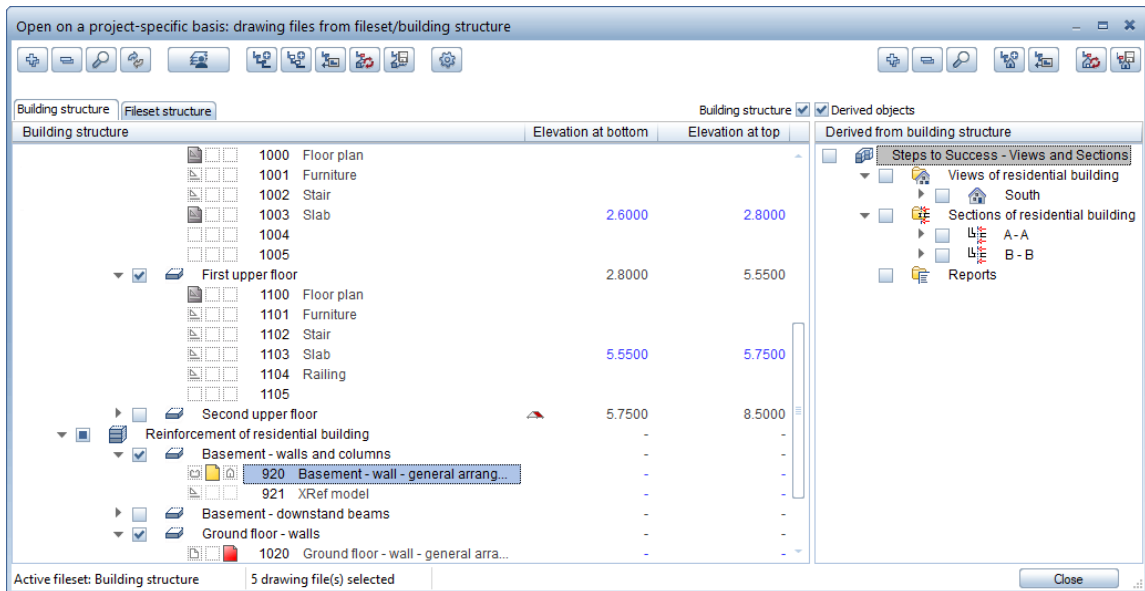
To create and delete additional links

- 1 Open the dialog box for selecting drawing files. Open the shortcut menu of drawing file **1020** and select **List reference drawing files** to check the links.

As you can see, this drawing file has links to the drawing files **1000**, **1003**, **1100**, and to itself.



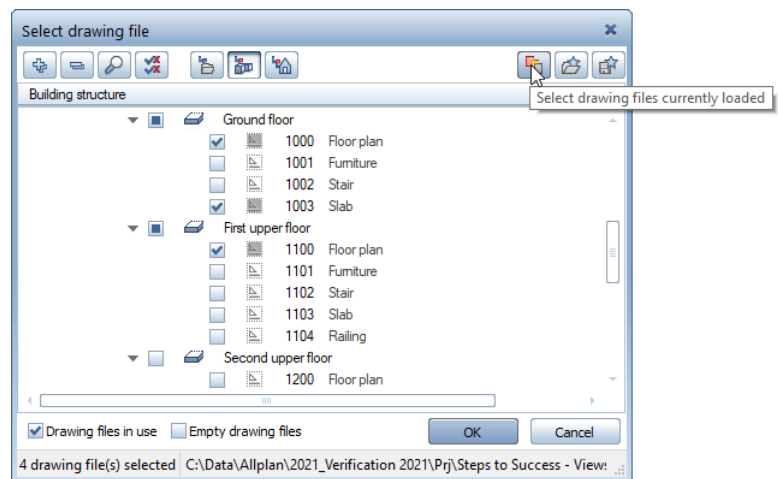
- 2 Open the shortcut menu of drawing file **1020** again, select **Set reference drawing files to reference mode**, and delete the contents of the drawing file, but do not delete the name of the drawing file.
- 3 Open the **Basement – walls and columns** structural level. Open drawing file **920** in **edit mode** and click **Close** to close the dialog box.



- 4 Use the clipboard to copy the entire contents of drawing file 920 to drawing file 1020. To do this, select CTRL+A, then CTRL+C, and finally CTRL+ALT+V.
- 5 Open the dialog box for selecting drawing files and close drawing file **920**.
- 6 Use **Move Linked** to move everything by 2.95 m in the z-direction.

The views and sections still have the link to drawing file 920 because you copied them by using the clipboard. Therefore, the wall elevation displays the mesh reinforcement of the basement. In addition, Allplan has incremented the mark numbers of the reinforcement.

- 7 Use  **Modify View, Section** to link all sections with the architectural data of the ground floor. To do this, click  **Select drawing files currently loaded** in the upper-right area of the **Select drawing file** dialog box.



The drawing files with the appropriate architectural data are already selected in reference mode in the building structure. Therefore, you do not need to select any additional drawing files.

- 8 Right-click any reinforcement element, select  **Rearrange Marks** on the shortcut menu, clear the **Lock** option, and click **OK** to confirm the dialog box.

Reinforcement based on an auxiliary 3D object

This training project uses a balcony, which is a simple, cube-like solid. If, however, you have a complicated component, which you do not want to model in detail, only a 2D general arrangement drawing is available to you for entering reinforcement.

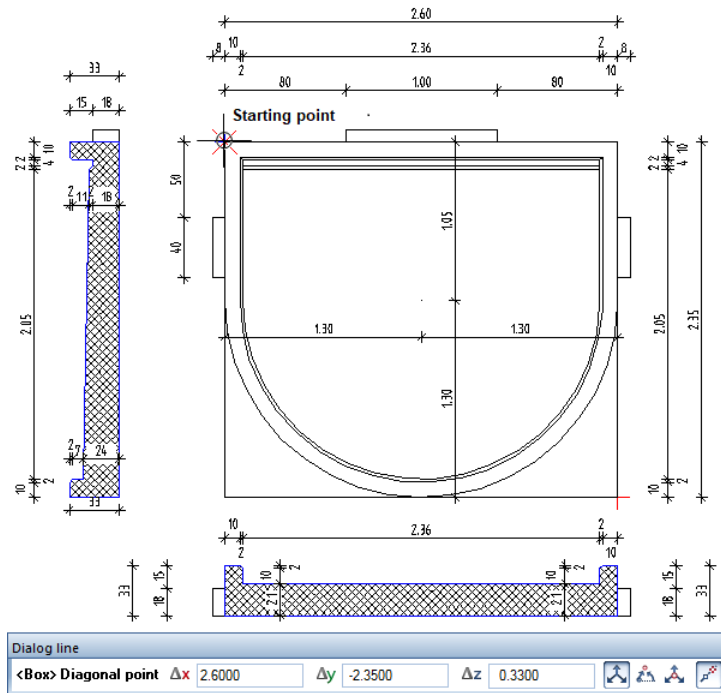
If you enter the first reinforcement element in a 2D drawing, Allplan cannot define the spatial orientation of the reinforcement. Therefore, you must select an existing view. But there is no view because you are about to create the first reinforcement element. So, you will use an auxiliary 3D solid instead.

Place an auxiliary box with the maximum dimensions on top of the 2D floor plan, create sections of this box, and place these sections on top of the 2D sections.

To create an auxiliary solid for the 2D general arrangement

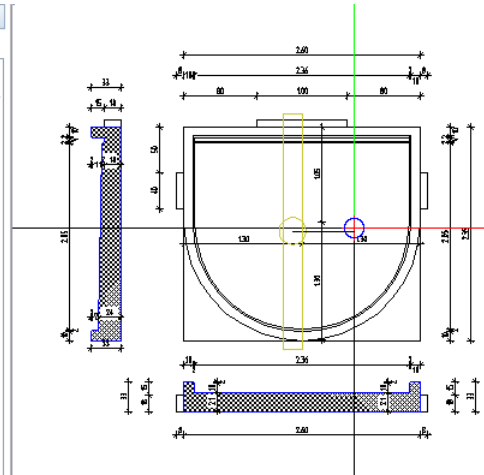
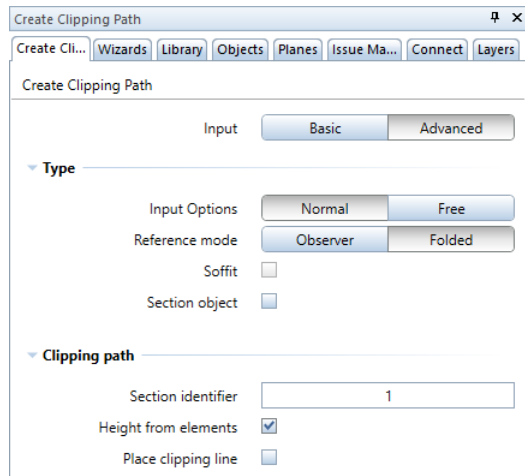
- 1 Go to the building structure on the left side, open the **Reinforcement of residential building – Ground floor – precast elements** structural levels, and select drawing file **1050**.
- 2 Open the dialog box or palette for layers, select the **List layers used in open documents** option, and make all layers  **Modifiable**.
- 3 Use the  **Box** tool (**Actionbar – Modeling** task – **3D Objects** task area) to create a box of 2.60 x 2.35 x 0.33 m that is parallel to the coordinate planes. These values (length x width x height) are the maximum dimensions of the precast element without Isokörbe.
 - Check that the **AR_GEN** layer is current.
 - To define the **starting point**, click the upper-left corner of the precast element in plan view.
 - To define the diagonal point, enter  **2.60**,  **-2.35**, and  **0.33**.

- Move the box by 2.60 m in the z-direction.



- 4 Open the Repeat drop-down list and click Generate Section. You can see the Create Clipping Path palette. Go to the Clipping path area and click the Set button to the right of Representation.
- 5 You can see the Clipping Path palette. Select the Section Identifier in construction-line format option and click Close to return to the Create Clipping Path palette.
- 6 In the Create Clipping Path palette, change the reference mode to Folded and clear the Section object and Place clipping line check boxes.

- 7 Select the **Height from elements** option. Define the clipping area so that it passes through the middle of the precast element and extends to the left. Define the viewing direction by clicking to the right of the circle.

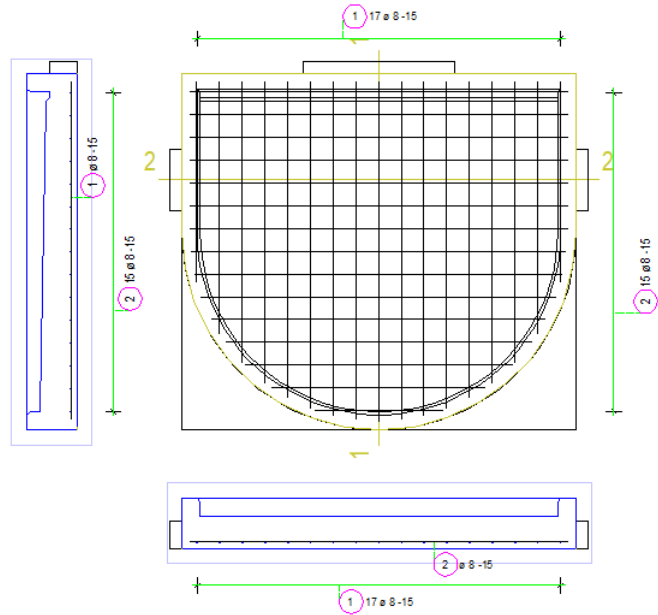


- 8 Check the settings. Clear the **Heading** check box and place the section so that it is congruent to the 2D section.
- 9 Create another section through the lateral *Isokörbe*; define the viewing direction from below.

Tip: Instead of sections, you can also create views of the box.

To do this, use  **Generate View** and define the viewing direction from the front or right.

- 10 Hide the **SH_SHL** layer for the dimensions and the **SU_HATCH** layer for the hatching. Create two-way area reinforcement at the bottom. Place **8-mm** bars spaced at **15 cm**.



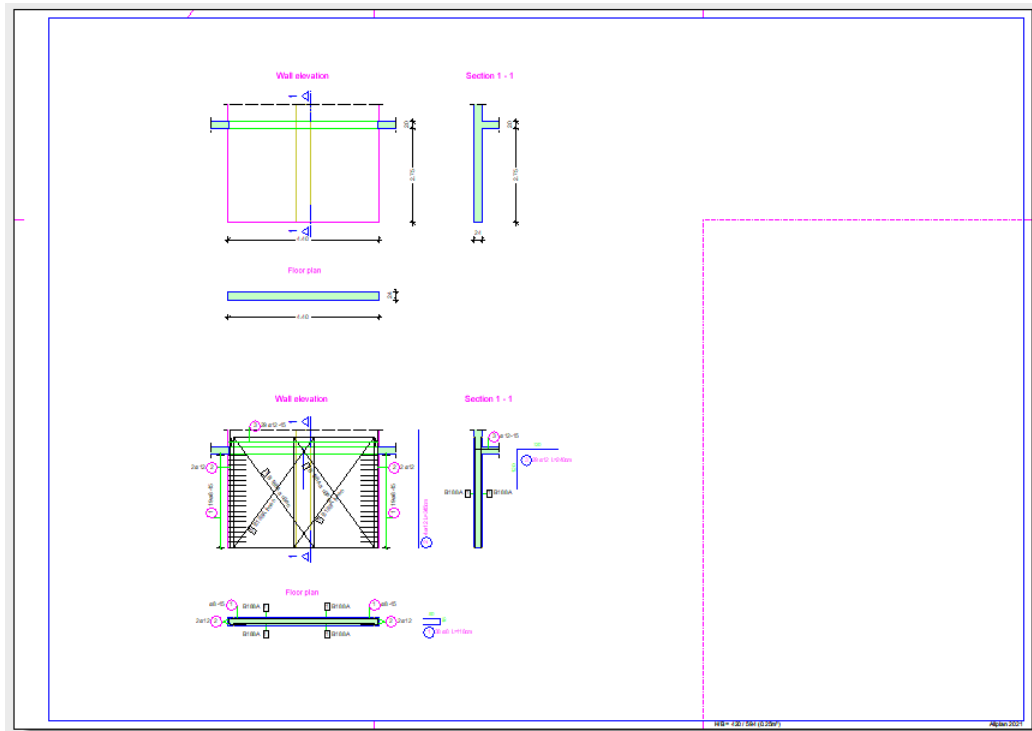
- 11 Before you output the layout, hide the **AR_GEN** layer of the auxiliary box to get the required reinforcement drawing.

Step 4: layouts

Create a layout for the interior wall in the first basement. You created the general arrangement and reinforcement in one drawing file. Consequently, the reinforcement model is also in the layout even though you do not need it for the layout. After you have placed the drawing file, you need to cut it so that the layout no longer contains the model data.

Note: Allplan always creates full schemas in the drawing file with the reinforcement model. Therefore, you need to place the drawing file with the reinforcement model in the layout even if the general arrangement and reinforcement are in different drawing files. If you created all data in a single drawing file, you need to use the following approach to create the general arrangement drawings and reinforcement drawings because the building model is also available.

Objective

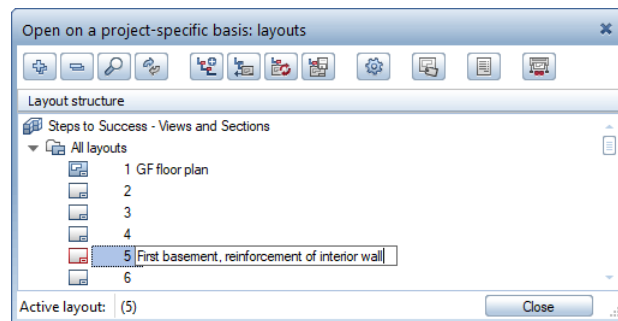


Assembling layouts

To assemble the layout for the wall, you first define the page format and border and then select the layout elements.

To define the page format and border

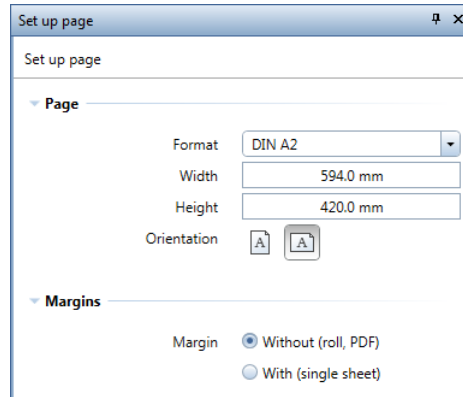
- 1 Select the **Layout Editor** task on the **Actionbar**.
- 2 Click  **Open on a Project-Specific Basis** (Quick Access Toolbar).
- 3 Click layout **5**, select the F2 key, enter **Reinforcement of interior wall, first basement** for the name of the layout, and click **Close** to confirm the dialog box.



- 4 Go to the **Actionbar** – **Layout Editor** task area and click  **Set Up Page**.

Tip: With this setting for the margins, Allplan always places the page so that its lower-left corner coincides with the lower-left corner of the printable area of the printer selected in the  **Print Layouts** tool. This ensures that the printout includes all elements that extend as far as the margins of the page.

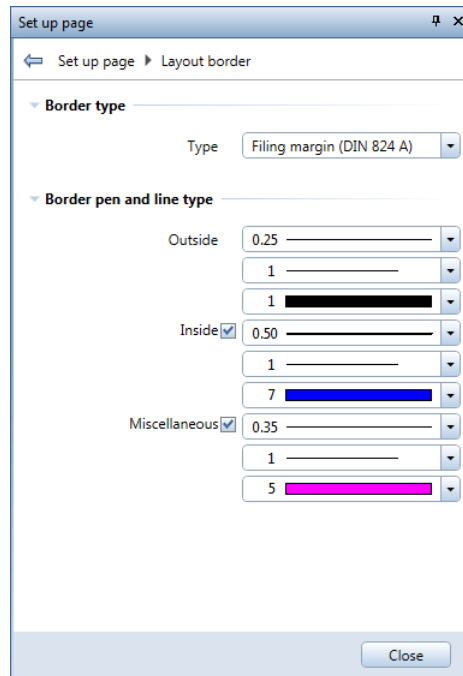
- 5 Define the format and its orientation in the **Page** area. In addition, specify the type and size of the **Margins**.



The 'Set up page' dialog box is shown with the 'Page' section expanded. It contains the following settings:

- Format:** DIN A2
- Width:** 594.0 mm
- Height:** 420.0 mm
- Orientation:** Portrait (indicated by the 'A' icon)
- Margins:**
 - Margin:** ☒ Without (roll, PDF) ☐ With (single sheet)

- 6 Go to the **Properties** area, click the button next to **Layout border**, and define the following parameters. Click **Close** to return to the **Set Up Page** palette.



The 'Set up page' dialog box is shown with the 'Layout border' section expanded. It contains the following settings:

- Border type:**
 - Type:** Filing margin (DIN 824 A)
- Border pen and line type:**
 - Outside:**
 - Line width: 0.25
 - Pen: 1 (black)
 - Inside:** ☒
 - Line width: 0.50
 - Pen: 7 (blue)
 - Miscellaneous:** ☒
 - Line width: 0.35
 - Pen: 5 (magenta)

Close

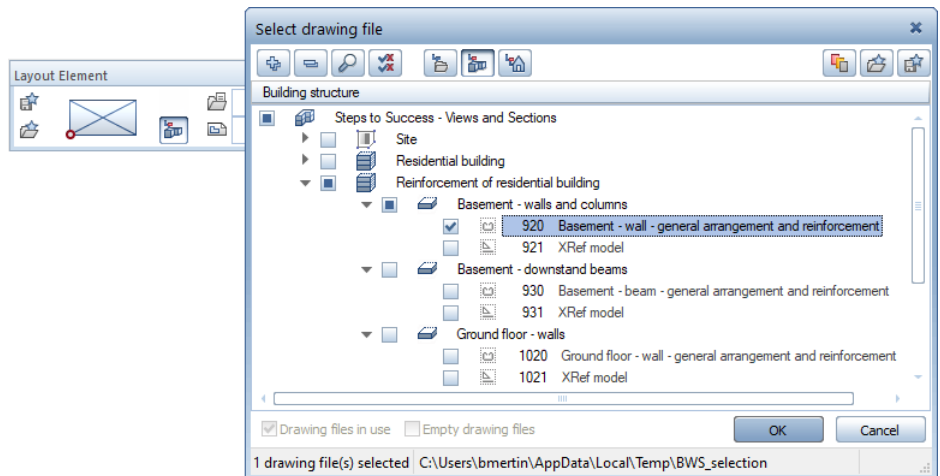
The layout border matches the size of the page. When you change the page, the size of the layout border adapts automatically.

Note: If you want to place a layout border of any size, use the  **Layout Border** tool (**Layout Editor** task area). You can use this tool to define custom border sizes by means of the input options.

- Click the button next to **Title block** to define a layout legend or a label style as a fixed property of the page.
Click **Close** to close the **Set Up Page** tool.

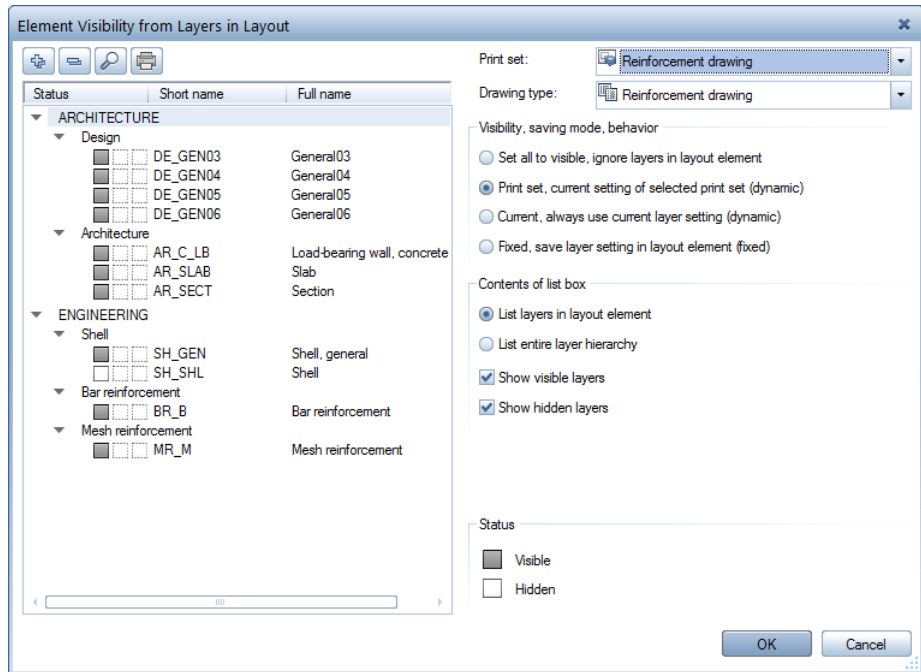
To place layout elements

- Click  **Layout Element** (**Layout Editor** task area).
- Click  **Building structure** on the **Layout Element** context toolbar and click  **Building structure** in the **Select drawing file** dialog box.



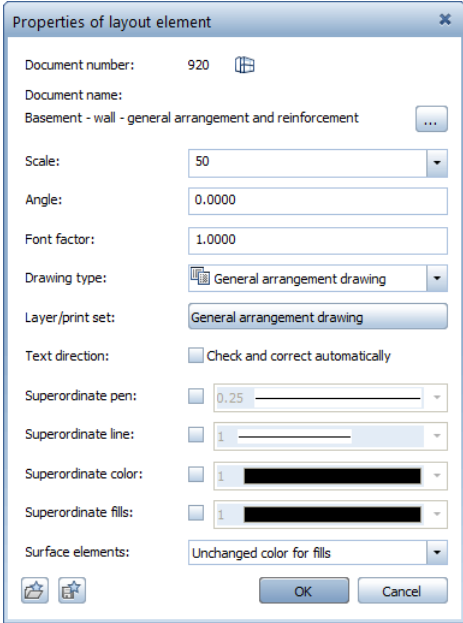
- Select drawing file **920** and click **OK** to confirm the dialog box.

- 4 Click in the **Layer/print set** box and select **Reinforcement drawing** for the drawing type. In addition, select the **Print set, current setting of selected print set (dynamic)** and **List layers in layout element** in layout element settings.
- 5 Select the **Reinforcement drawing** print set and click **OK** to confirm the dialog box.



- 6 Place the selected drawing file in the layout so that the model data is outside the layout border.  **Move (Edit task area)** the layout element if this is necessary.
- 7 Select ESC to finish selecting layout elements.
- 8  **Copy (Edit task area)** the drawing files downward in alignment.
- 9 Select ESC to close the tool and double-click the original drawing file.

- 10 Select **General arrangement drawing** for both the drawing type and the print set. Click **OK** to confirm the dialog box.



The image shows a dialog box titled "Properties of layout element". It contains the following fields and options:

- Document number: 920
- Document name: Basement - wall - general arrangement and reinforcement
- Scale: 50
- Angle: 0.0000
- Font factor: 1.0000
- Drawing type: General arrangement drawing
- Layer/print set: General arrangement drawing
- Text direction: ☐ Check and correct automatically
- Superordinate pen: 0.25
- Superordinate line: 1
- Superordinate color: 1
- Superordinate fills: 1
- Surface elements: Unchanged color for fills

At the bottom, there are icons for a folder and a star, and buttons for "OK" and "Cancel".

Layout without model data

Elements that are outside the layout border are not included in printouts. However, these elements appear in print preview. You can use a layout section or layout window to hide the model data.

To create a layout section

- 1 Go to the **Actionbar**, expand the **Crop Layout** task area, and click  **Convert Layout Section**.
- 2 Select layout **6** and enter **50** for both the **Scale** and **Ref LS** (reference layout scale).

Lay #	6	F Fctr	Layout	Angle	0.00
Scale	50	Ref LS	50	Layout section	static

- 3 Click an element of the layout border.

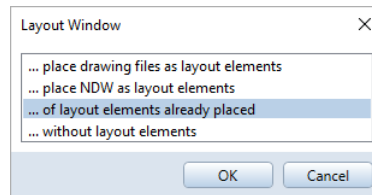
A message tells you that the section has been created.

Note: To create sections of smaller areas of the layout, use  **Layout Border** to place a layout border of the required size and click this border. Thus, you can also cut large layouts into smaller "sublayouts".

- 4 Click  **Load Layout Section or Layout** (**Crop Layout** task area).
 - 5 Click **LayNam** in the input options and select layout **6**.
 - 6 Move the entire layout so that the lower-left corner of the layout border coincides with the lower-left corner of the page.
 - 7 Click  **Load Original Layout** (**Crop Layout** task area).
-

To create a layout window

- 1 Go to the **Actionbar – Layout Editor** task area and click  **Layout Window**.
- 2 Click ... of layout elements already placed.

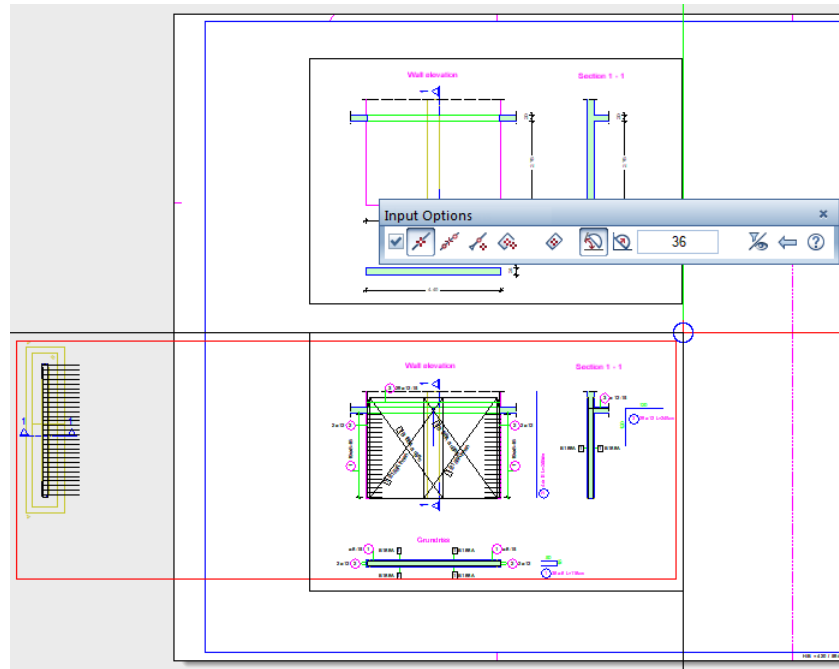


- 3 Select the layout element at the top (general arrangement drawing).

Note: You must create two separate layout windows from the general arrangement drawing and the reinforcement drawing because Allplan automatically places all elements in the layout window so that they are congruent.

- 4 To define the size of the layout window, click two diagonal points (lower-left and upper-right points) so that the reinforcement model is outside the layout window.

- 5 Repeat this procedure for the layout element at the bottom (reinforcement drawing).



- 6 Select ESC twice to finish entering the layout window and to close the tool.

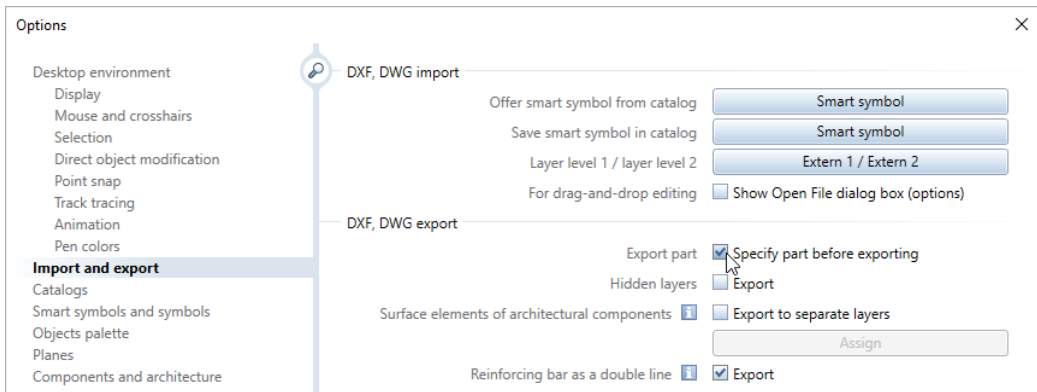
Step 5: exchanging data

You can exchange data both in document edit mode and in layout edit mode. The export options vary depending on the mode. When importing data, make sure you select the correct layout scale.

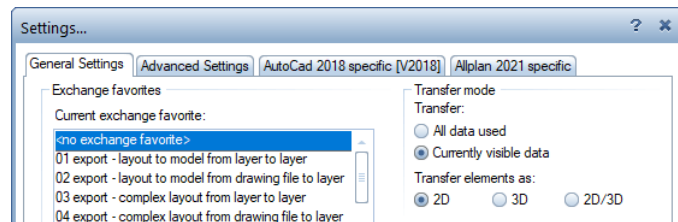
DWG export of drawing files

You do not need to export the model data unless it is part of the general arrangement drawings and reinforcement drawings. Change the settings in the options for the interfaces so that you can select only the associative views and sections.

To do this, click  **Options** ( **Default Settings** drop-down list on the Quick Access Toolbar) and select **Import and export**. Select the **Export part** check box.



Basically, the elements that are visible on the screen should be the elements that you want to transfer. Go to the **Transfer mode** area of the configuration settings and specify that you want to transfer the **Currently visible data** as **2D** elements.

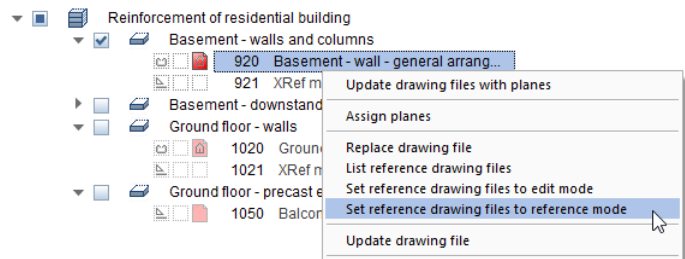


Note: If you do not want to transfer the view borders of the associative views and sections, open the **Advanced Settings** tab and make sure the **not printable elements** entry is not selected in the **Transfer elements** area.

To transfer the data of the associative views and sections, you do not need to load the drawing files with the model data. As with layouts, you need to load the drawing file with the reinforcement model because this drawing file contains the full schemas.

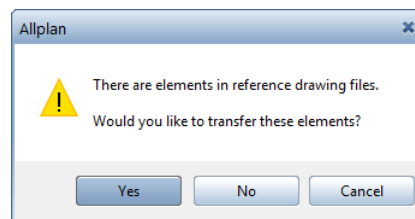
- 1 Double-click in the workspace. Go to the **Basement - walls and columns** structural level and double-click drawing file **920**.

Note: If you created the general arrangement and reinforcement in different drawing files, open the drawing file with the reinforcement model in  **edit** or  **reference mode**. Alternatively, open the shortcut menu of the drawing file with the associative views and sections and select **Set reference drawing files to reference mode** so that you do not interrupt the workflow in a workgroup environment.



- 2 Open the drop-down list of the  Allplan icon, point to **Export**, and click **Export AutoCAD Data...**
Click **Yes** if the drawing file with the reinforcement model is open in reinforcement mode.

Note: If the drawing files contain layers that are  **visible, frozen**, you can choose to export them too.

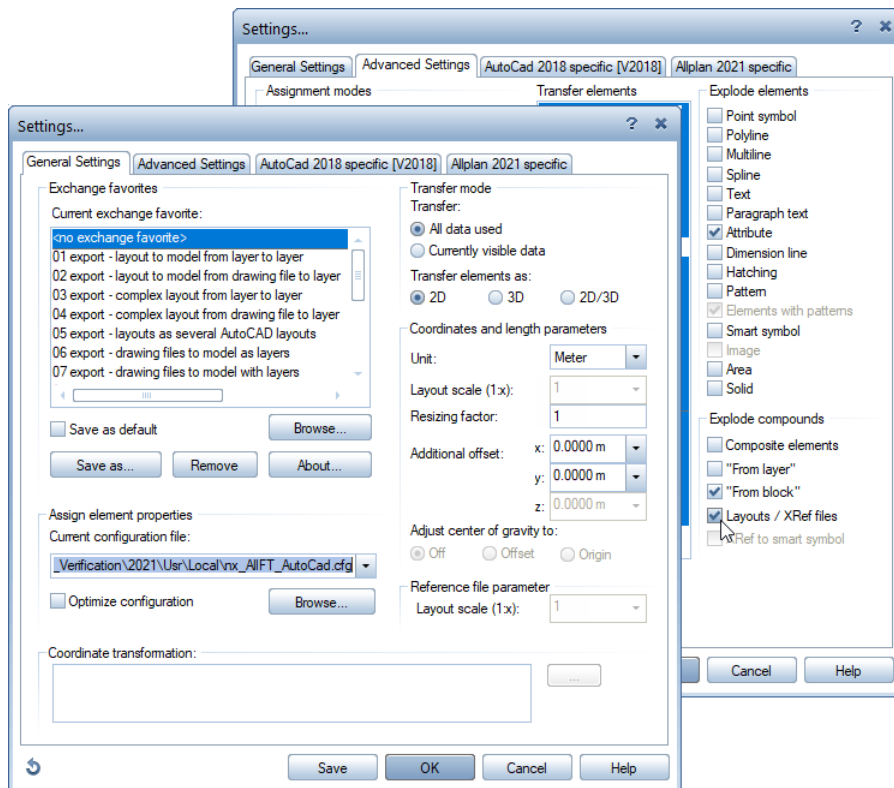


- 3 Select the associative views and sections, define the settings for the export, and export the data.

DWG export of layouts

When exporting layouts, make sure all the layers that are visible in the layout elements are  **Modifiable**. Otherwise, you get layers without access rights when you import the data.

Go to the **Transfer mode** area (**General Settings** tab) of the configuration settings and specify that you want to transfer the **Currently visible data** as 2D elements. Open the **Advanced Settings** tab and select the **Layouts / XRef files** option in the **Explode compounds** area.



Appendix: working without links

If you do not need a building model that is always up to date, you can work without links. To create the general arrangement drawings and reinforcement drawings of the components, copy and convert the required part of the floor plan to a new, empty drawing file and create the required sections by means of the tools in the **Sections** task area.

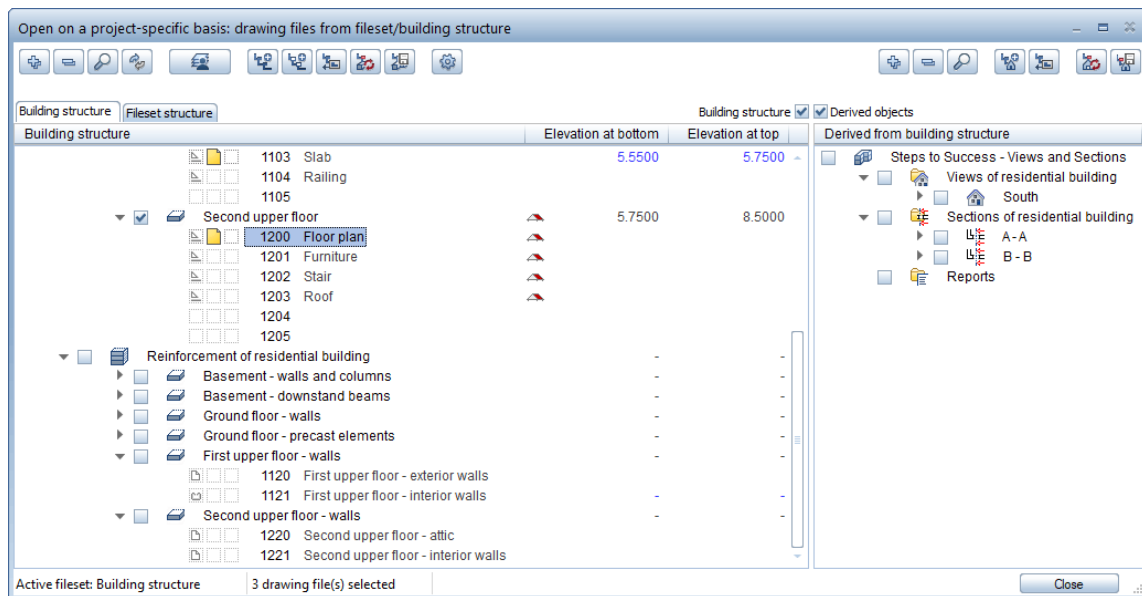
This approach not only produces versions of the general arrangement data and reinforcement data but also ensures that no user changes the building model inadvertently.

This approach is also useful if a lot of linked drawing files in a complex project slow down your computer. You can even work with and without links in one and the same project.

Organization in the building structure

If you want to use this approach, you need to make only some changes to the contents of the **building** structural levels for the general arrangement and reinforcement. Create a drawing file under each **building** structural level required and create the data of the components in this drawing file.

This training project comes with the required structure for the first and second upper floors.



Note: FC export considers the left side of the building structure only. If you do not want to use the IFC interface to transfer the general arrangement and reinforcement data, you can also create the required folders for the components on the right side of the building structure.

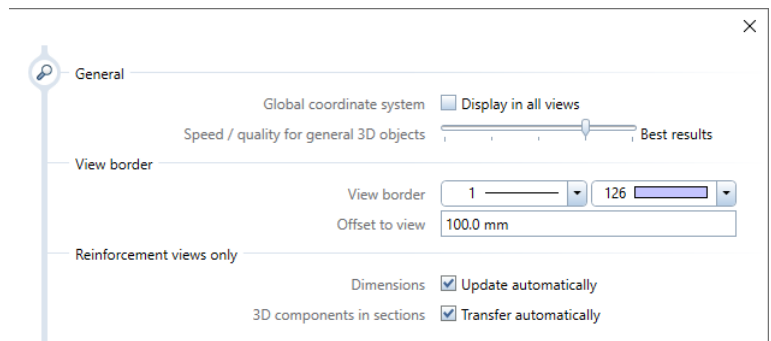
Defining view options

The settings in the options are the same, regardless of whether you work with or without linked drawing files. You do not produce linked drawing files as long as you create everything in the same drawing file.

Make sure that you do not link drawing files inadvertently by copying or moving data.

To check the options

- 1 Click  **Options** and then **Views**.
- 2 Check the settings. The **Transfer 3D components automatically to sections** option applies only to views and sections of the **Reinforcement Views** task area.

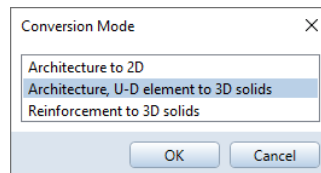


Copying and converting a part of the floor plan

By copying the model data, you make sure that all users in a workgroup environment can work smoothly without interruption. Check and update the building data before you create the final layout.

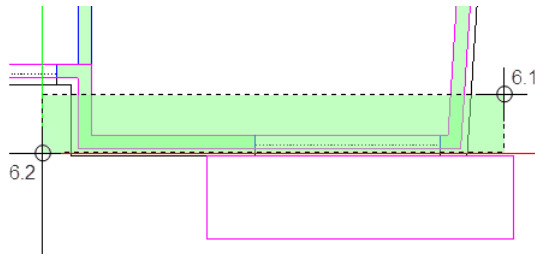
To copy and convert the model data for the exterior wall

- 1 Double-click in the workspace. Go to the left side of the **Building structure**, open the **Residential building – First upper floor** structural levels, and double-click drawing file **1100**.
- 2 Double-click in the workspace again. Open the **Second upper floor** structural level and change the status of the drawing files **1103** and **1200** to  **edit mode**. This automatically selects the **Second upper floor** structural level.
- 3 **Close** the dialog box and select the **General arrangement drawing** layer print set.
- 4 Go to the **Change** task area, open the flyout menu of the  **Convert Elements** tool, click  **Copy, Convert Elements Across Drawing Files**, and select the **Architecture, U-D Element to 3D solids** conversion mode.



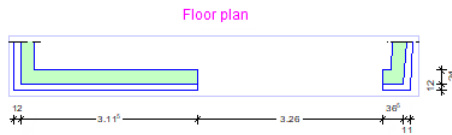
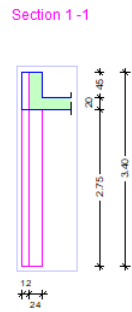
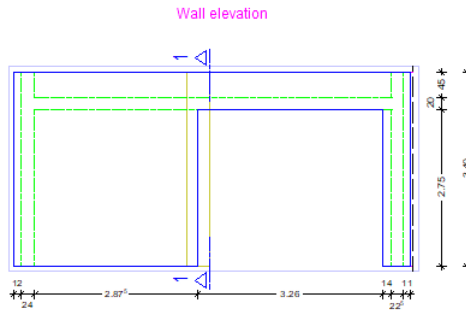
- 5 You can see the **Select destination drawing file** dialog box. Click  **Building structure**, go to the **Reinforcement of residential building – First upper floor – walls** structural levels, select the check box of drawing file **1120**, and click **OK** to confirm the dialog box.

- 6 *Select elements to copy and convert:* Enclose the lower-right exterior wall in a selection rectangle that you open from right to left. Make sure that the slab above the balcony is not selected ( **Select Elements based on Direction** is selected in the **Work Environment** task area; the selection rectangle is shown as dashed lines, and the area it covers is highlighted in light green).



- 7 Double-click drawing file **1120** and change the scale on the status bar to **1:50**.
- 8 Select **SH_GEN** for the current layer. Use the  **Generate Section** tool to create a section of the lower exterior wall. Select the **Observer** reference mode and define the viewing direction from below.
- Change the **Section Identifier** to **A** and select the **Height from elements** option. Define the clipping area in the floor plan so that the lower end of the section is within the thermal insulation.
 - Open the **Label** palette, enter **Wall elevation** for the heading, clear the **Place section identifier** check box, and place the section next to the floor plan.
- 9 Create another section through the wall and a section of the floor plan.

- 10 Adjust the extent of the section object of section 1 - 1 so that the thermal insulation is completely visible. Move the headings and use  **Dimension Line** to create the associative dimensions on the SH_SHL layer.



Using changed model data

If you create empty views and sections by deleting the model data, the view borders and clipping paths remain unchanged.

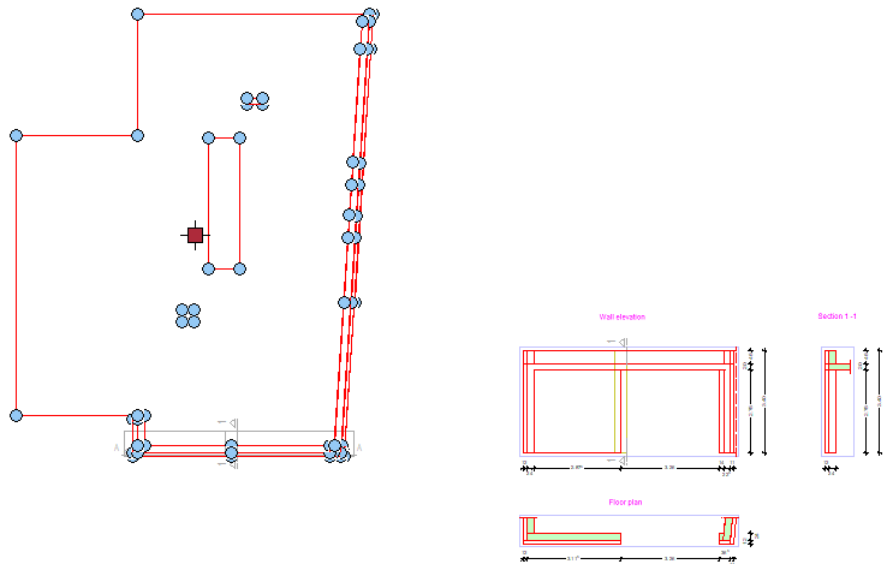
Consequently, you can simply delete the model data, regardless of whether the views and sections contain only general arrangement data or also reinforcement data.

Tip: If you want to keep the model data for the time being so that you can check everything, assign a new layer to the entire floor plan. Make sure this layer has not yet been used in the drawing file. Then copy the changed model data and delete the original floor plan afterward.

To use changed model data

- Drawing file **1120** is open. The **General arrangement drawing** print set is selected.
- 1 Switch the **AR_SECT** layer to  **Visible, frozen** so that you do not delete the clipping paths together with the floor plan.
- 2 Select the entire floor plan and click  **Delete** (shortcut menu or **Edit** task area).

Note: You can also use  **Filter Step by Step** (in the **Filter** task area) to filter specific elements. For example, you can use  **Filter by Layer** to filter the model data of the building again by clicking  **Apply to all elements, new filter** if the clipping paths are on the **DEFAULT** layer.



- 3 Open the dialog box for selecting drawing files and turn off the **First upper floor – walls** structural level in the **Reinforcement of residential building** structural level. Select the **Residential building** structural level and the **First upper floor** and **Second upper floor** structural levels.
- 4 Use  **Copy and Convert Elements across Drawing Files** to copy and convert the model data for the exterior wall in drawing files **1100**, **1103**, and **1200** to drawing file **1120**. Follow the steps described in "Copying and converting a part of the floor plan". A message tells you that the selected drawing file is in use. Click **Yes** to confirm.
- 5 Switch back to drawing file **1120**.

The associative sections automatically contain the copied model data.
- 6 Make the **AR_SECT** layer  **Modifiable** and create the associative dimensions.

Using reinforcement again

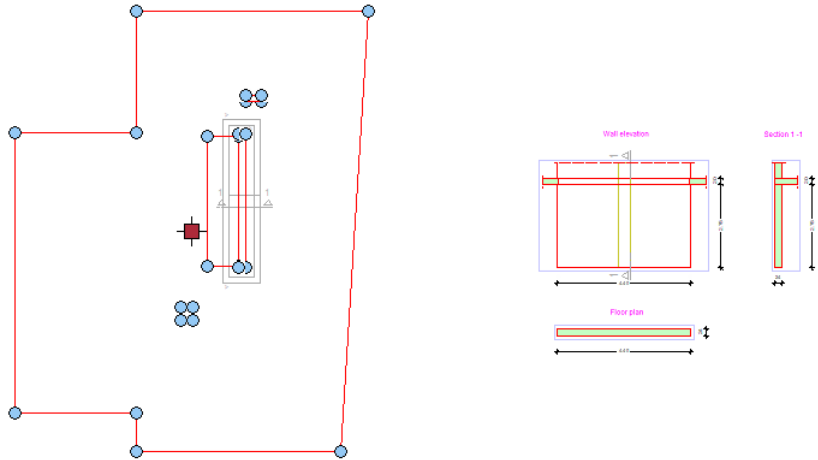
You can find the reinforced wall next to the interior stair on the first upper floor in drawing file **1121** in the training project.

The process of using reinforcement again is similar to using changed model data.

To use the wall reinforcement of the first upper floor for the second upper floor

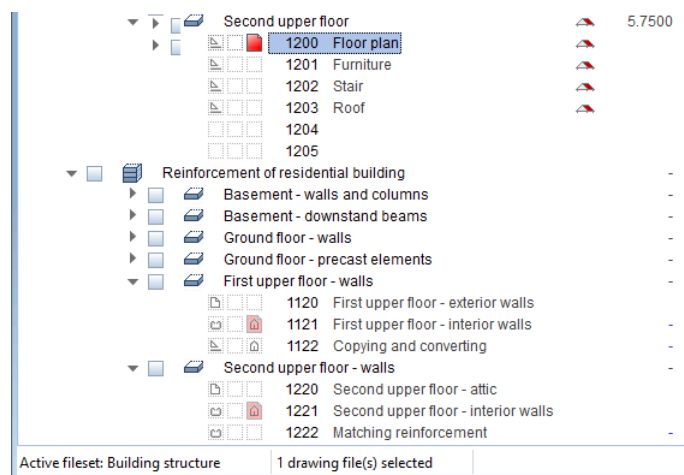
- 1 Open drawing file **1121** and close all the other drawing files. Check the reinforcement by selecting the **Reinforcement drawing** print set.
- 2 Double-click in the workspace to open the dialog box for selecting drawing files. As described in step 3 (see "Using reinforcement again" on page 50), open the shortcut menu of drawing file **1121** and **Copy** the entire contents of the drawing file to drawing file **1221**. To insert the contents, open the shortcut menu of the **Second upper floor – walls** structural level and select **Paste under**.
- 3 Rename drawing file 1221 **Second upper floor – interior walls** and double-click drawing file **1221**.
- 4 Go to the **Actionbar – Sections** task area and click  **Move Linked**. Use CTRL+A to move the entire floor plan with the model data of the building and the reinforcement model so that the reinforcement model is in the correct position for the new component that you want to reinforce. In this example, the wall is congruent on the next floor. Therefore, move the floor plan by 2.95 m (= story height) in the z-direction.

- 5 Select the **General arrangement drawing** print set, switch the **AR_SECT** layer to  **Visible, frozen**, and delete the model data of the building from the floor plan as described.



In this case, you do not create empty sections by deleting the model data. Therefore, the associative dimensions remain unchanged.

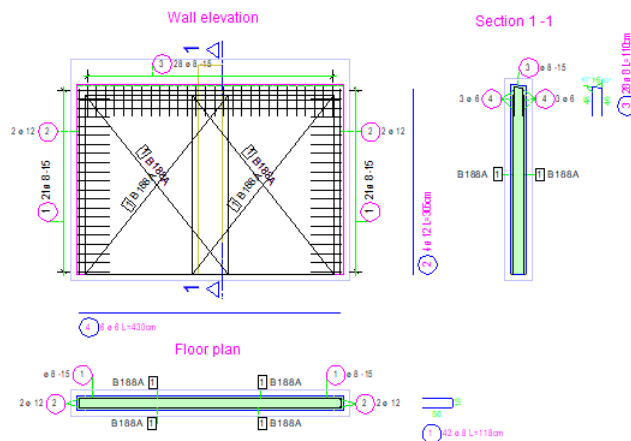
- 6 Go to the left side of the building structure and select drawing file **1200** in the **Second upper floor** structural level.



- 7 Use  **Copy, Convert Elements Across Drawing Files** to copy and convert the model data for the wall next to the interior stair to drawing file **1221**.
A message tells you that the selected drawing file is in use. Click **Yes** to confirm.
- 8 Switch back to drawing file **1221** and select the **All layers visible** print set.

The associative sections automatically contain the copied model data.
- 9 Adjust the associative dimensions of the sections and select the **Reinforcement drawing** print set.
- 10 Delete the L-shaped bar. Select  **Stretch Entities** and shorten the meshes by 30 cm and the longitudinal bars by 40 cm. Use **direct object modification** or the **Properties palette** to enlarge the placing regions of the stirrups. Select  **Bar Shape** and create a new open stirrup at the top and additional cross bars. Then rearrange the reinforcement.

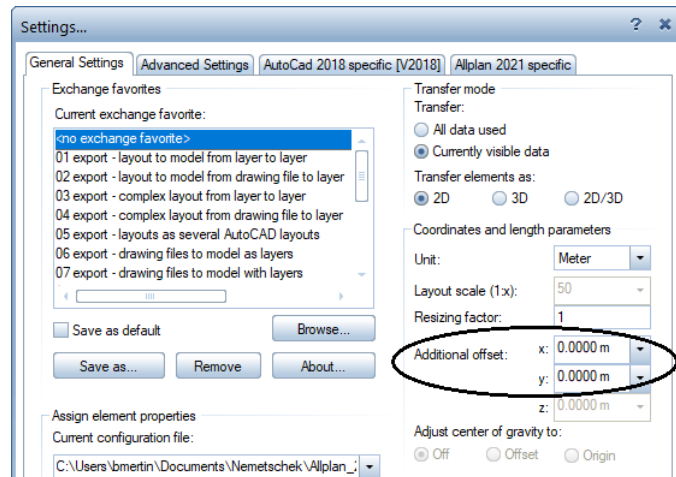
The result might look like this:



Data exchange

When working without drawing file references, you can export drawing files as described (see step 5 (see "Step 5: exchanging data" on page 73)). As the model data and the views and sections are in a single drawing file, you do not need to select additional drawing files. So that you can only select the associative views and sections, make sure the corresponding setting is selected in the  **Options**.

If you have created a section of the floor plan and you want to place the data imported in the correct position in the building model, move the section of the floor plan and all the associated sections by fixed x/y values, which are based on the building model. Then enter these values (make sure that you use the opposite sign) for the additional offset in the export file.



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